Applied Value
Building Products Report

Industry Report

Q4 2020

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Applied Value Building Products Report

Executive Summary & Introduction

Financial Benchmarking

Industry Trends

Applied Value Presentation

Introduction

Applied Value's building products practice

Building products is a core practice area for Applied Value. We have supported a wide range of clients across the building products and construction value chain, and we own and operate two building products companies.

We generate client value and provide tangible results by applying an unbiased perspective, having a hands-on approach, and recognizing the importance of delivering value fast.

About this report

This report tracks the financial performance of major players in the US building materials and products industry and covers sub-sectors across the full building products sphere, from raw materials companies to manufacturers of doors and pipes.

We hope you find this report insightful and we welcome feedback and opportunities for further discussion.

Disclaimer

This report is based on the benchmarked companies' Q4 2020 earnings reports, performance beyond Q4 2020 is not considered. However, due to the current situation in several underlying raw materials price markets, in particular lumber and steel, we are making callouts throughout the report addressing notable price considerations going into Q1 2021.

This report analyzes the financial performance of 36 building materials and products companies operating in the US, and covers 434 BUSD in revenue

Building materials companies

Segment	Company	Revenue (LTM BUSD)	Country
Cement, aggregates and ready-mix concrete	CRH	27.6	Ireland
	Lafarge Holcim	26.2	France
	Heidelberg Cement	22.0	Germany
	CEMEX	13.0	Mexico
	Taiheiyo Cement	8.5	Japan
	Vulcan	4.9	US
	Martin Marietta	4.4	US
Glass	Saint Gobain	46.6	France
	AGC	13.7	Japan
	Owens Corning	7.1	US
Lumber and wood	Weyerhaueser	7.5	US
	UFP	5.2	US
	West Fraser	4.6	Canada
	Canfor	4.3	Canada
	Resolute	2.8	US
Steel	Arcelor Mittal	53.3	Netherlands
	Nucor	20.1	US
Bricks and masonry	Boral	4.2	Australia
	Wienerberger	4.1	Austria

106.1 BUSD 67.4 BUSD 24.4 BUSD 73.4 BUSD 8.3 BUSD

The report covers 279.5 BUSD of building materials revenue

Building products companies

Segment	Company	Revenue (LTM BUSD)	Country
Building envelope, roofing, siding flooring and insulation	Builders FirstSource	8.6	US
	Beacon	7.1	US
	Kingspan	5.6	Ireland
	Cornerstone	4.6	US
Doors and windows	LIXIL	12.9	Japan
	ASSA ABLOY	10.7	Sweden
	Sanwa	4.2	Japan
	JELD-WEN	4.2	US
	Masonite	2.3	US
Piping	Forterra	1.6	US
Kitchen and bath	Masco	7.2	US
	Fortune Brands	6.1	US
	Geberit	3.4	US
HVAC-R, fire and security	Daikin	23.6	Japan
	Johnson Controls	22.1	Ireland
	Carrier	17.5	US
	Trane	12.5	Ireland

25.9 BUSD 34.3 BUSD 1.6 BUSD 16.7 BUSD 75.6 BUSD

The report covers 154.1 BUSD of building products revenue

Source: Capital IQ

The benchmarked companies operate across a wide array of building materials and products sub-sectors

Building materials companies

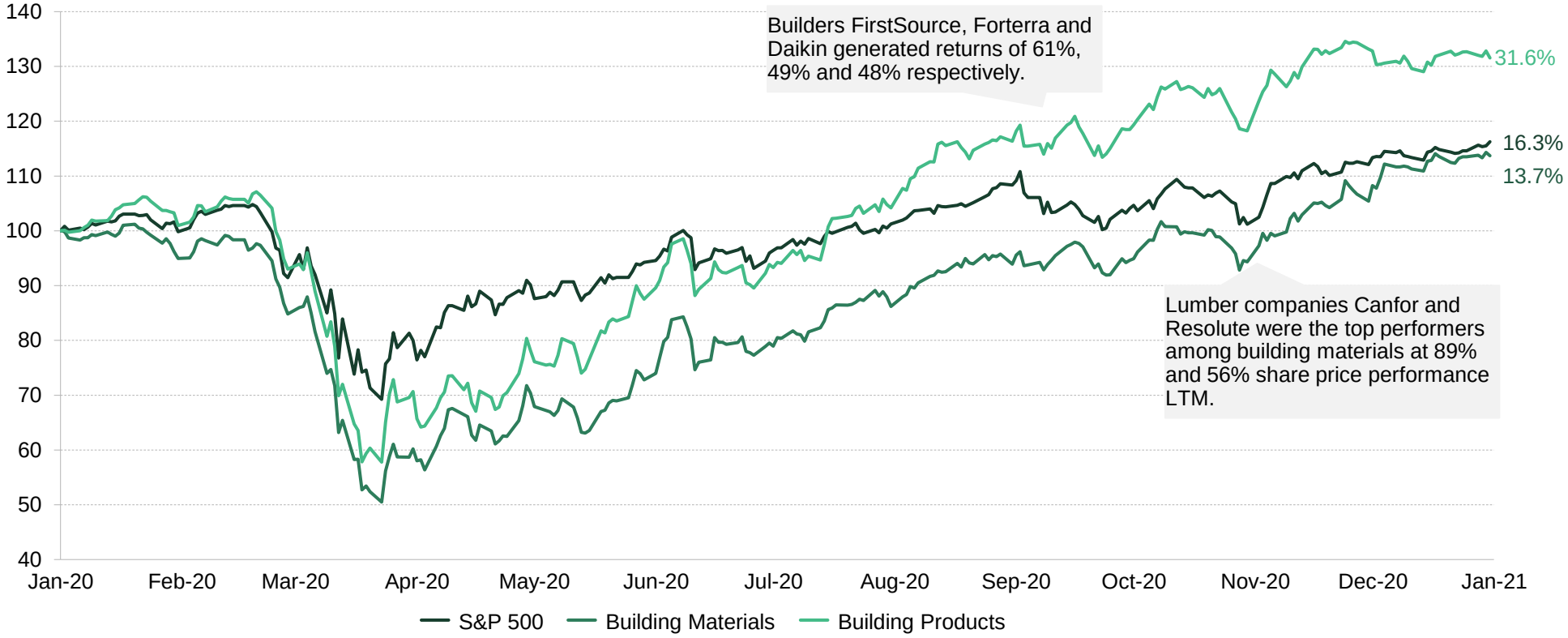
Company	Description
CRH	Cement, aggregates, concrete and asphalt manufacturer
Lafarge Holcim	Manufacturer of cement, aggregates and ready-mix
Heidelberg Cement	Producer of cement, aggregates and asphalt
CEMEX	Produces cement, aggregates, ready-mix and clinker
Taiheiyo Cement	Global provider of cement
Vulcan	Producer of construction aggregates
Martin Marietta	Supplier of cement, aggregates and concrete
Saint Gobain	Manufacturer of glass and other construction materials
AGC	Global glass manufacturer
Owens Corning	The largest manufacturer of fiberglass composites
Weyerhaueser	Manufacturer of wood products
UFP	Manufacturer of wood and wood-alternative products
West Fraser	Producer of lumber, plywood and other wood products
Canfor	Producer of sustainable wood-building solutions
Resolute	Provider of wood and other forest products
Arcelor Mittal	Operates steel manufacturing and mining facilities
Nucor	Manufactures steel and steel products
Boral	Supplier of bricks and other construction materials
Wienerberger	The world's largest brick producer

Building products companies

Company	Description
Builders FirstSource	Manufacturer & supplier of various building products
Beacon	Supplier of roofing and other building products
Kingspan	Flooring, panels and insulation manufacturer
Cornerstone	Provides roofing, siding, insulation and more products
LIXIL	Window, door, external and interior products producer
ASSA ABLOY	Door and security solutions provider
Sanwa	Manufacturer of doors, windows and shutters
JELD-WEN	Producer of windows and doors
Masonite	Designs and manufactures interior and exterior doors
Forterra	Manufacturer of water and drainage pipes
Masco	Kitchen and bath products conglomerate
Fortune Brands	Portfolio of various kitchen and bath companies
Geberit	Manufacturer of various bath products
Daikin	Provider of air-conditioning solutions
Johnson Controls	Producer of HVAC, security and fire equipment
Carrier	HVAC-R solutions provider
Trane	Manufacturer of HVAC-R systems

Building products have outperformed the S&P500 LTM, Carrier was the best performing individual stock with a return of 214% following an April IPO

Indexed share price performance
1 Jan 2020=100



Source: Capital IQ
Notes:
1. Building products index includes all building products companies included in report
2. Building materials index includes all building materials companies included in report

Executive summary

Key takeaways

Lumber has outperformed

- Lumber companies have seen strong performance in Q4 2020 with an average revenue growth rate of 37% and significant increase in profitability.

Surge in raw materials prices

- Raw materials prices have been trending up since summer 2020, with lumber and steel seeing the largest price increases.
- The surging prices have had positive impact on revenue growth and profitability of the analyzed companies.

Strong demand

- Demand for building materials and products has been strong as the pandemic sparked a boom in new house construction and home renovations projects.
- The housing market has also seen strong demand and house prices have been rising across the US.

Mixed revenue performance

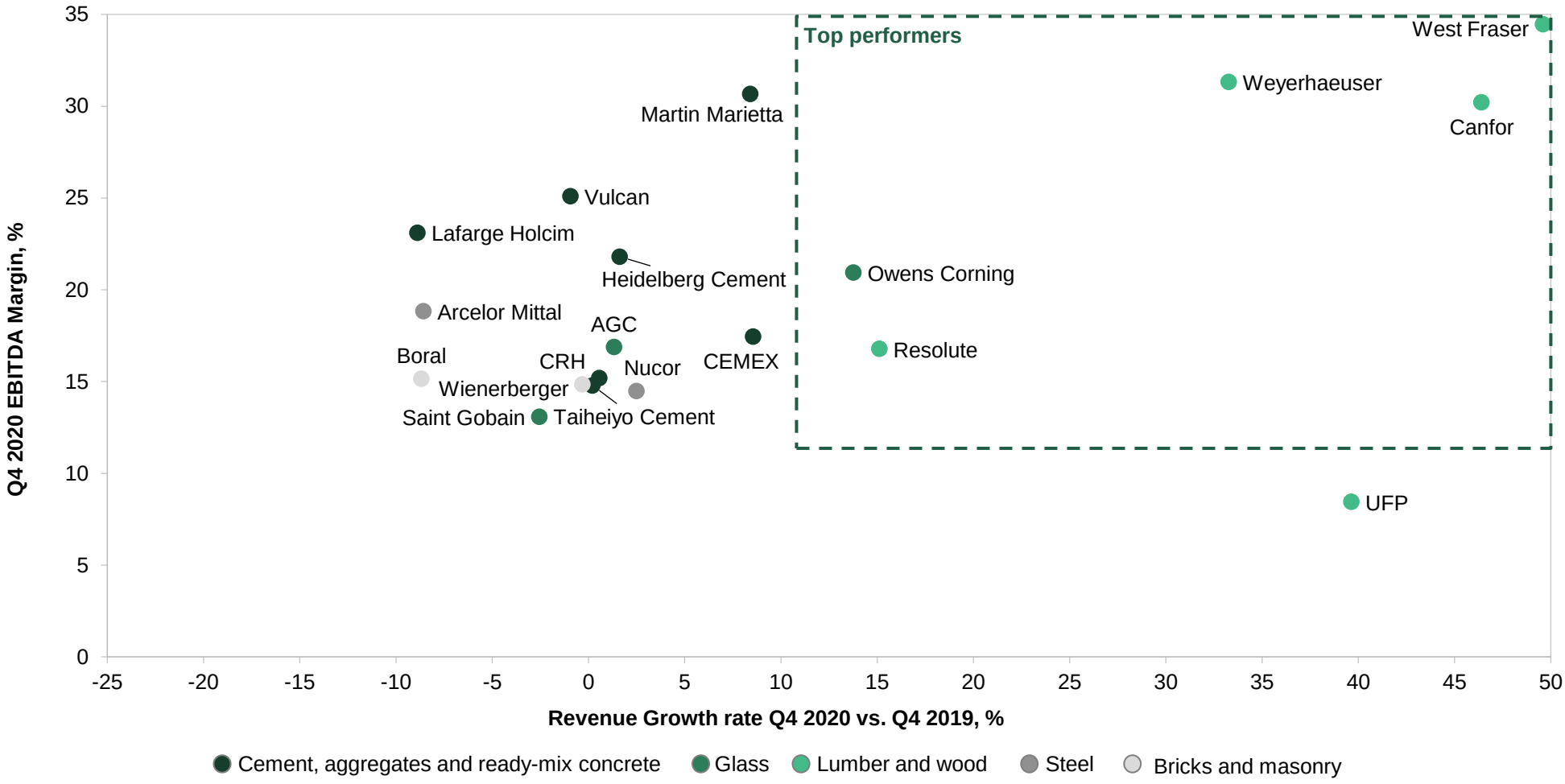
- The benchmarked players have seen mixed revenue growth with growth rates spanning from +50% to -20%.
- Segments that have seen direct effect of the rising raw materials prices have seen the strongest revenue performance.

Improved profitability

- The industry has seen improved profitability YoY, with lumber and steel companies seeing the largest improvements in EBITDA margin.

Lumber companies have outperformed other building materials sub-sectors YoY. The highest profitability is found in lumber and cement companies

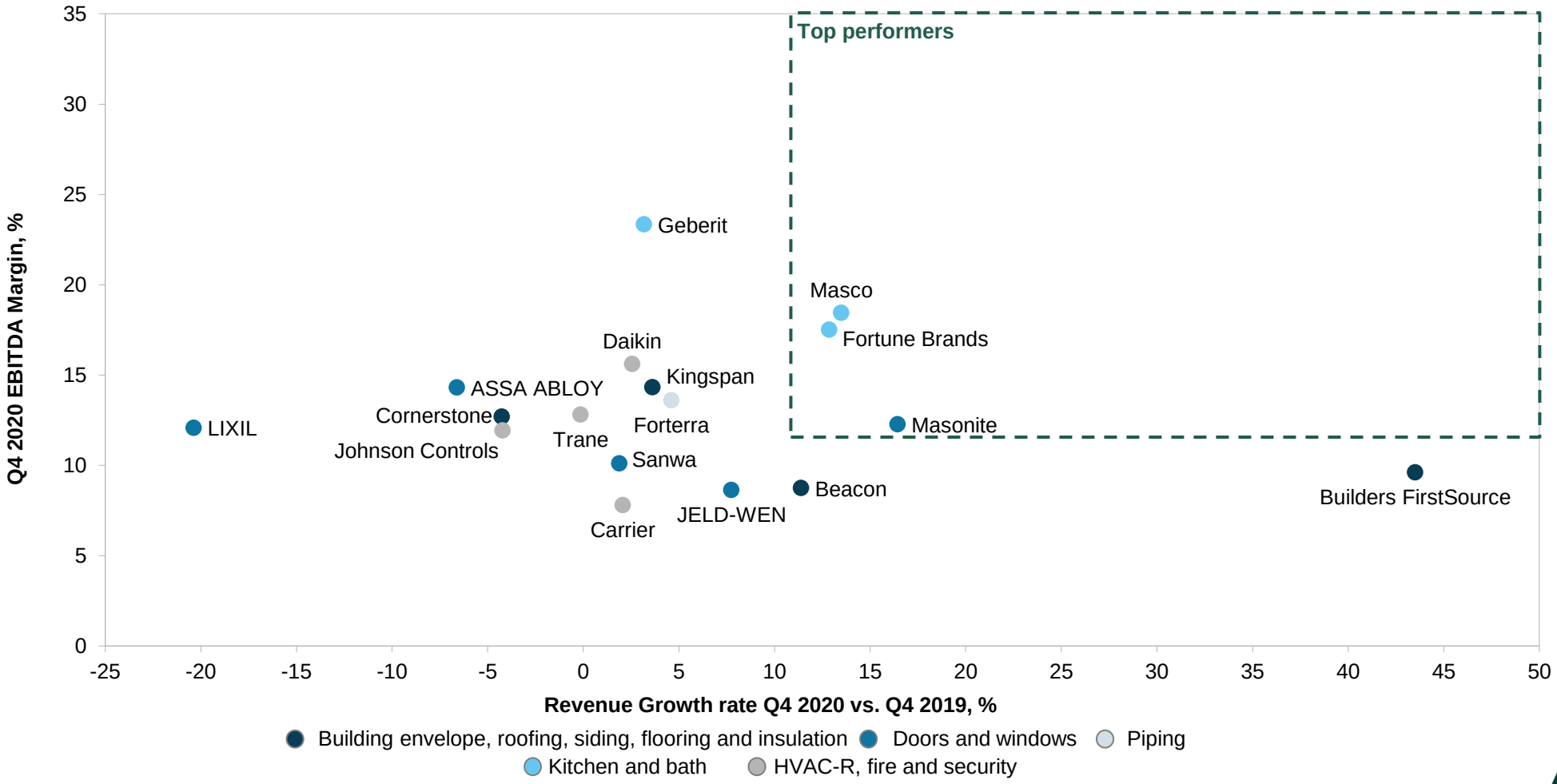
Building materials revenue growth rate and EBITDA margin



Source: Capital IQ, Applied Value analysis

Kitchen and bath companies have outperformed other sub-sectors on profitability in Q4, Builders FirstSource saw the strongest revenue growth

Building products revenue growth rate and EBITDA margin



Source: Capital IQ, Applied Value analysis

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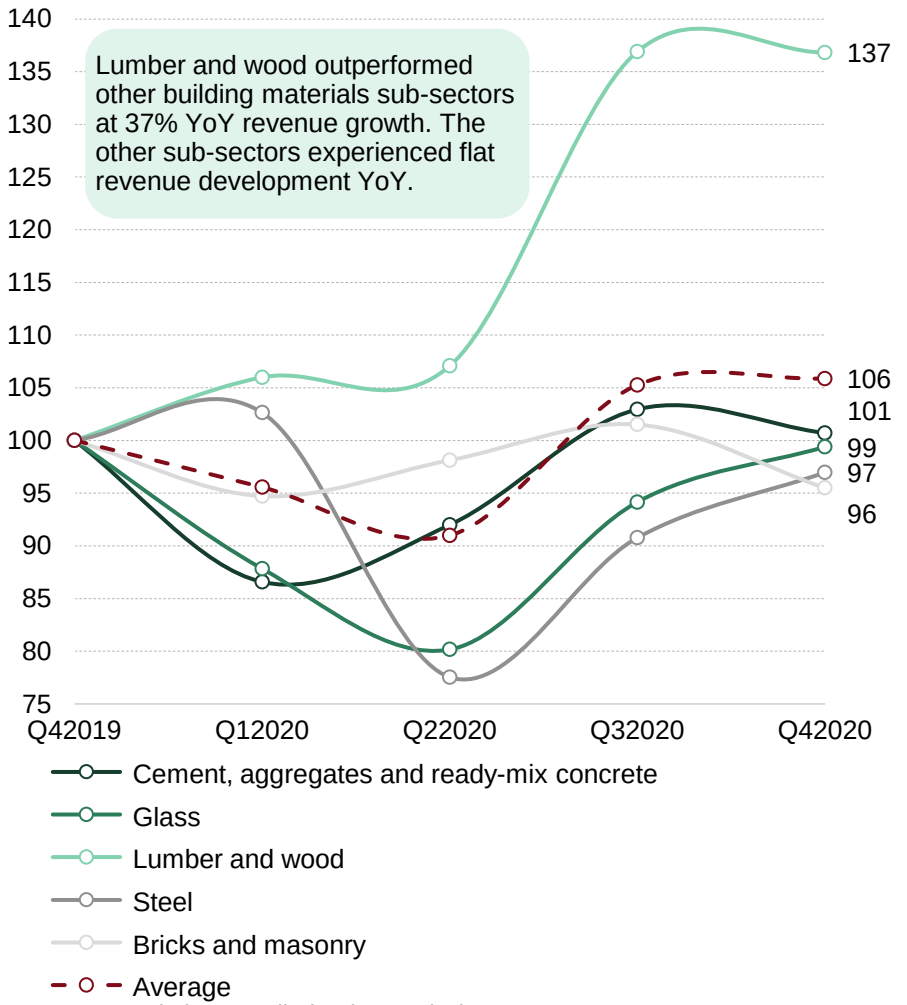
Key financial takeaways from the building materials and building products sectors

Revenue Growth	› The industry has seen overall strong revenue growth Q4'20 vs. Q4'19, with building materials and building products growth rates at 10% and 5% respectively. › Top performers in the building materials sector were lumber companies West Fraser, Canfor and UFP with YoY growth rates at 49.6%, 46.4% and 39.6%. The growth in revenue was driven by surging lumber prices during H2 2020. › The high price of lumber also had positive effect on Builders FirstSource which outperformed other building products companies at 43.5% revenue growth.
Profitability & Costs	› The rising prices of lumber and steel have had positive impact on lumber and steel companies' EBITDA margins and both sub-sectors saw significant improvement in profitability during Q4. The rising raw materials prices led to strong revenue growth while costs remained stable. › The benchmarked companies have managed to control costs, and the industry has seen a slight decrease in COGS over sales and SG&A over sales YoY.
Return on Invested Capital	› Building materials ROIC has been highly volatile L5Y and is highly affected by price movements in the underlying raw materials markets. › Building products companies have achieved on average higher ROIC than building materials companies LTM at 8.4% vs. 6.1% › Masco, Geberit and West Fraser are the top performers with ROIC of 25.6%, 21.7% and 18.3% respectively

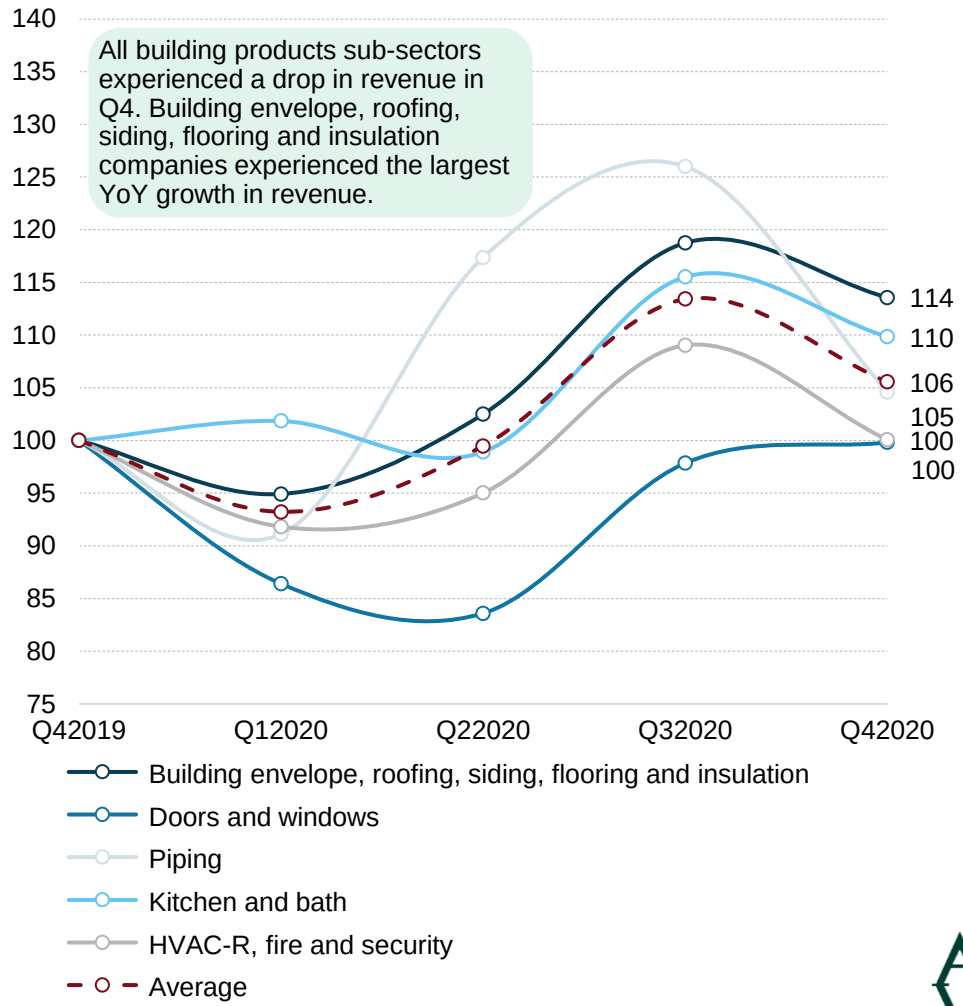
Lumber has seen the strongest revenue development during 2020 following tight supply and strong demand in the market

Building materials quarterly revenue development
Index Q4 2019 = 100



Source: Capital IQ, Applied Value analysis

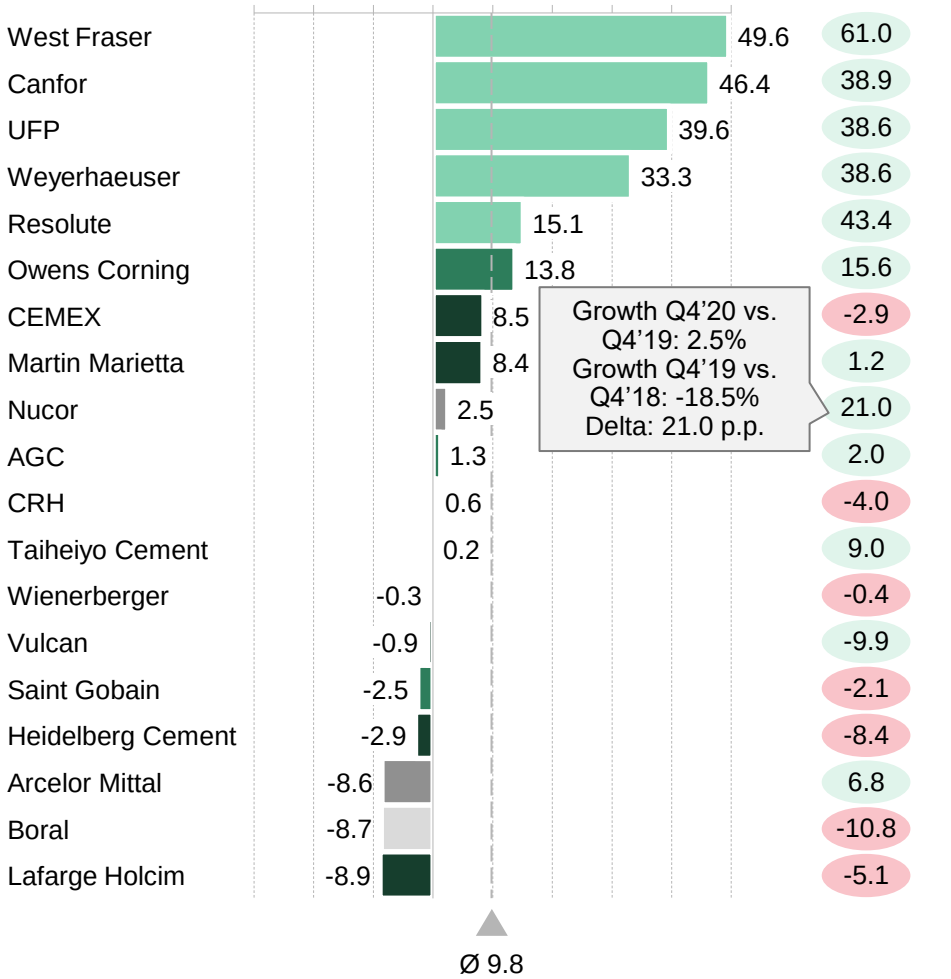
Building products quarterly revenue development
Index Q4 2019 = 100



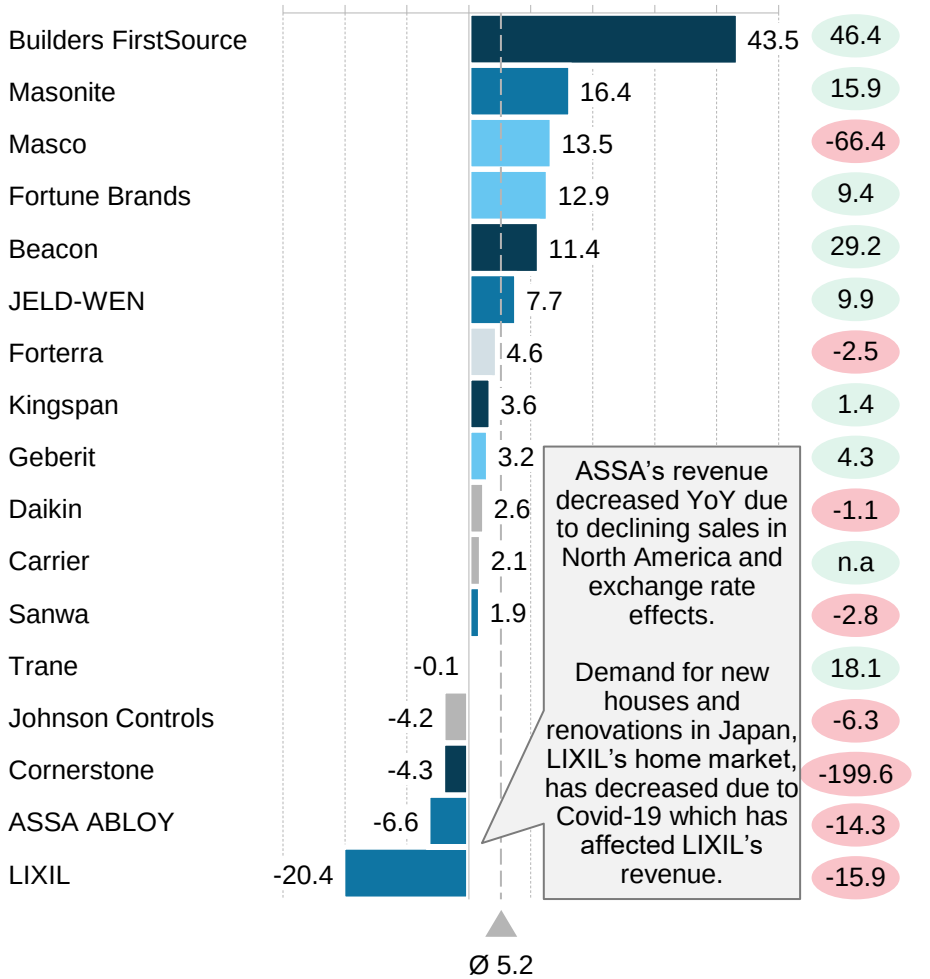
Lumber companies and Builders FirstSource have experienced significant increase in revenue YoY

Change in growth rate YoY Q4'20/Q4'19 vs. Q4'19/Q4'18

Building materials revenue growth YoY
Q4'20 vs. Q4'19, %



Building products revenue growth YoY
Q4'20 vs. Q4'19, %

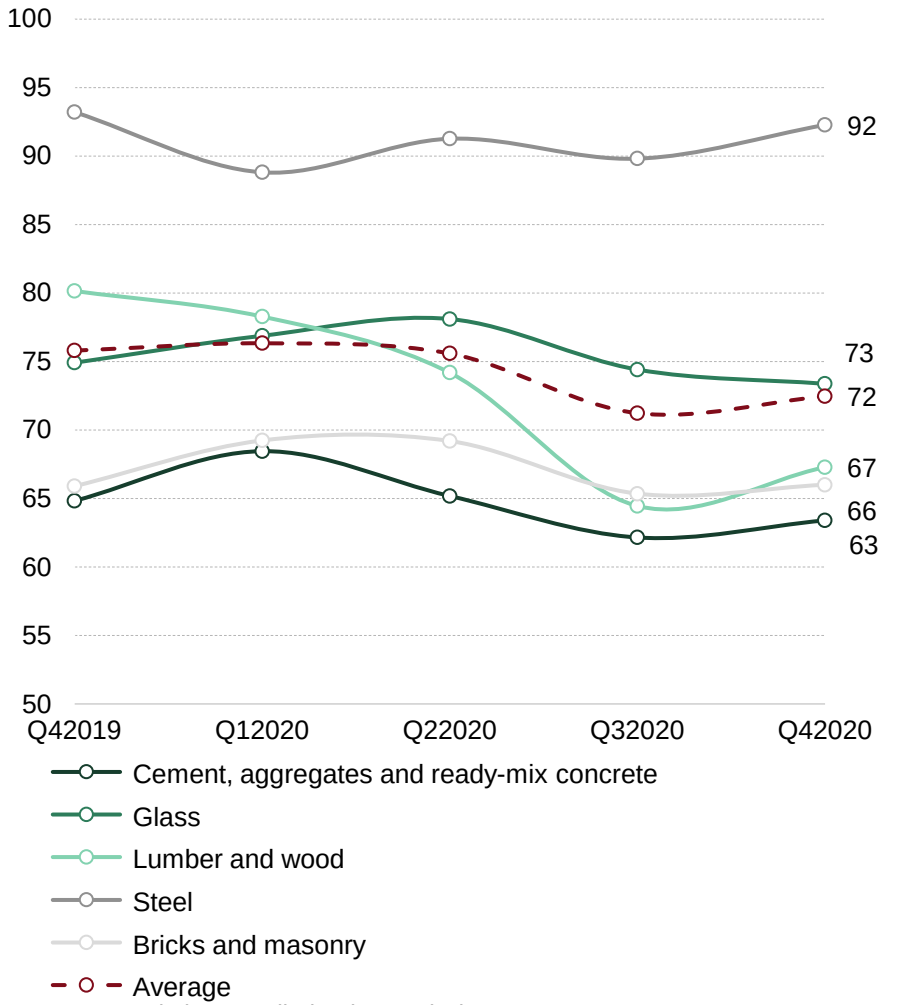


Cement, aggregates and ready-mix concrete
Glass
Lumber and wood
Steel
Bricks and masonry
Building envelope, roofing, siding, flooring and insulation
Doors and windows
Piping
Kitchen and bath
HVAC-R, fire and security

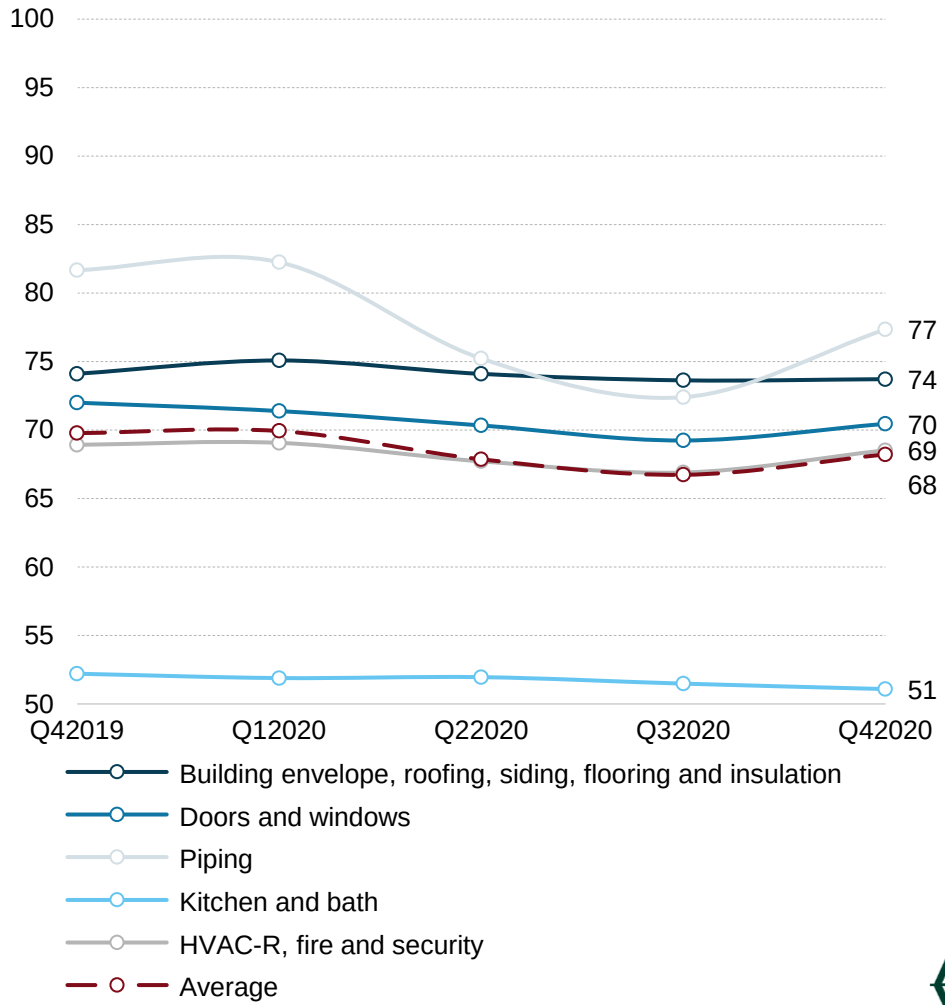
Source: Capital IQ, Applied Value analysis
Note: Heidelberg Cement data is based on half-year reporting

COGS/sales have remained rather stable YoY, kitchen and bath has the lowest while steel companies face the highest COGS/sales

Building materials quarterly COGS/sales development
%

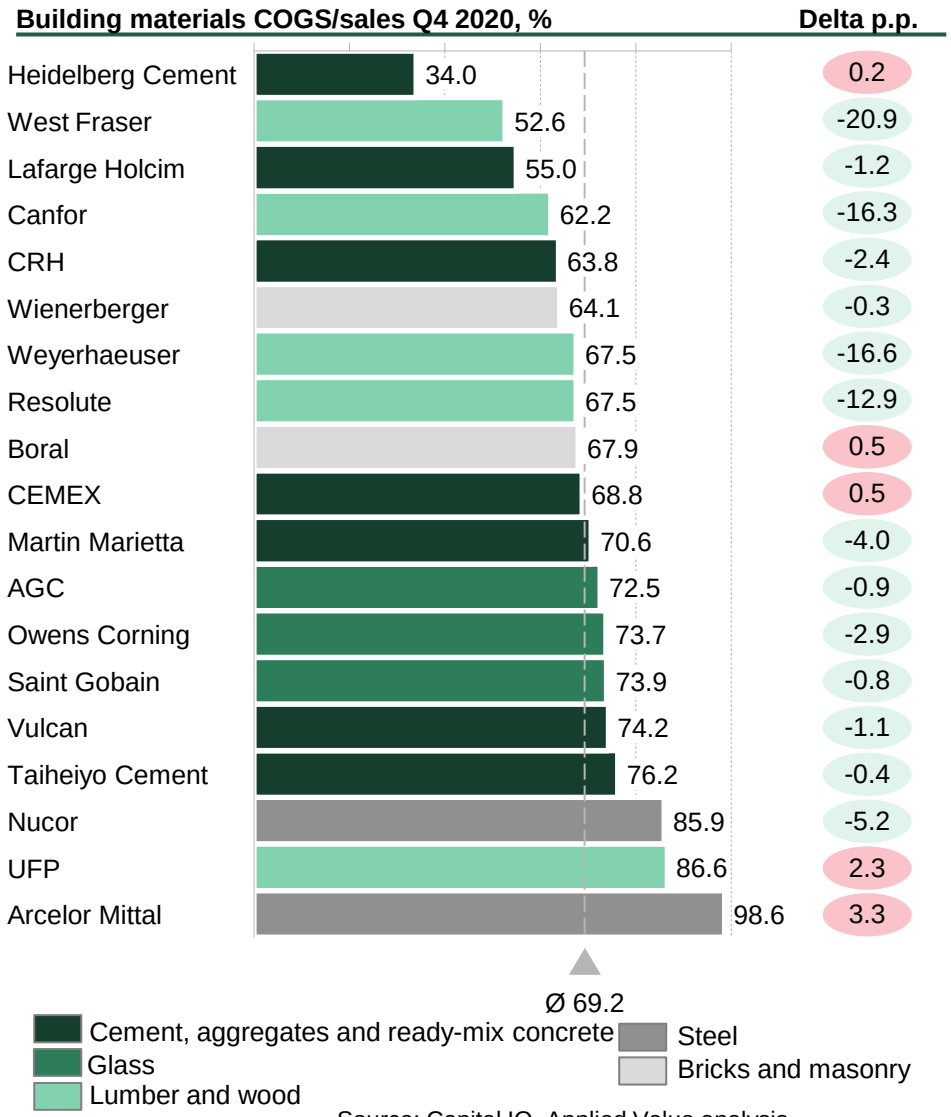


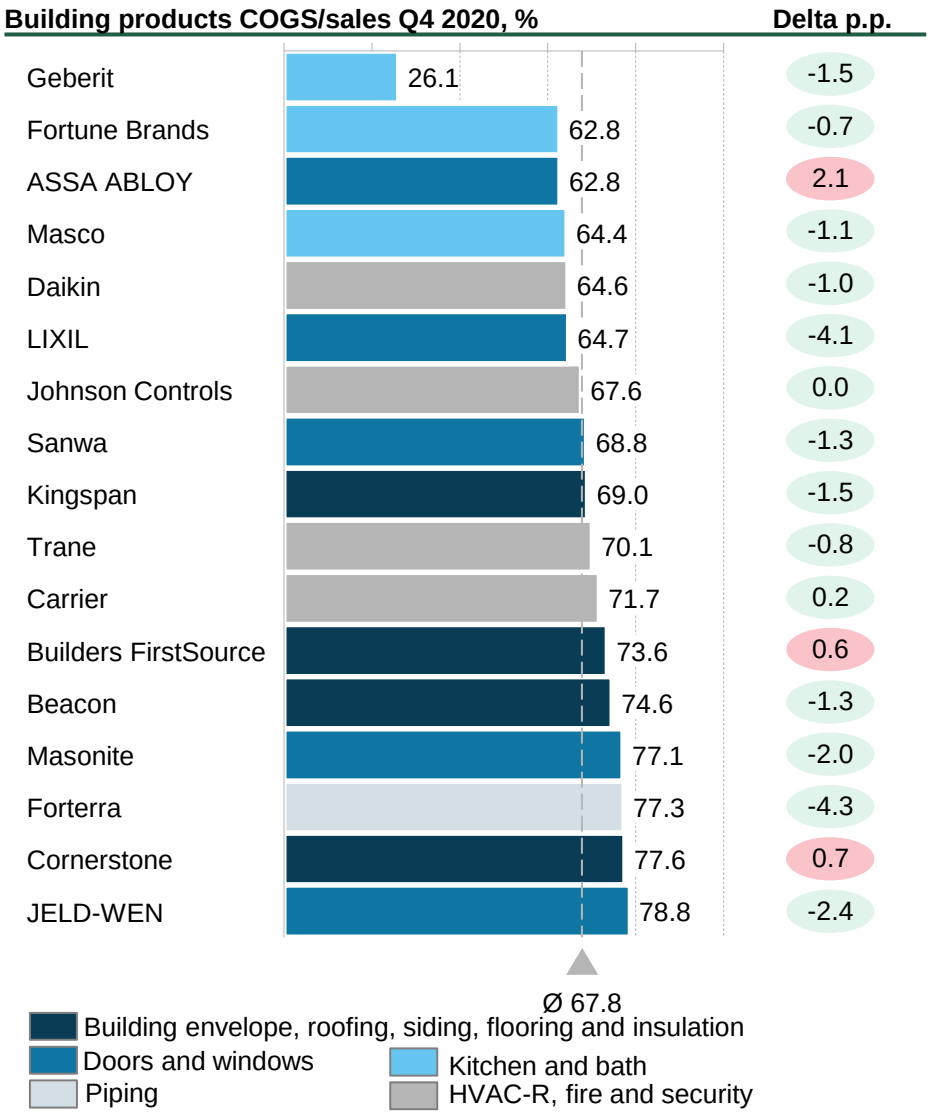
Building products quarterly COGS/sales development
%



Source: Capital IQ, Applied Value analysis

The benchmarked companies have seen overall positive development in COGS/sales, steel companies and UFP have the highest COGS/sales ratios

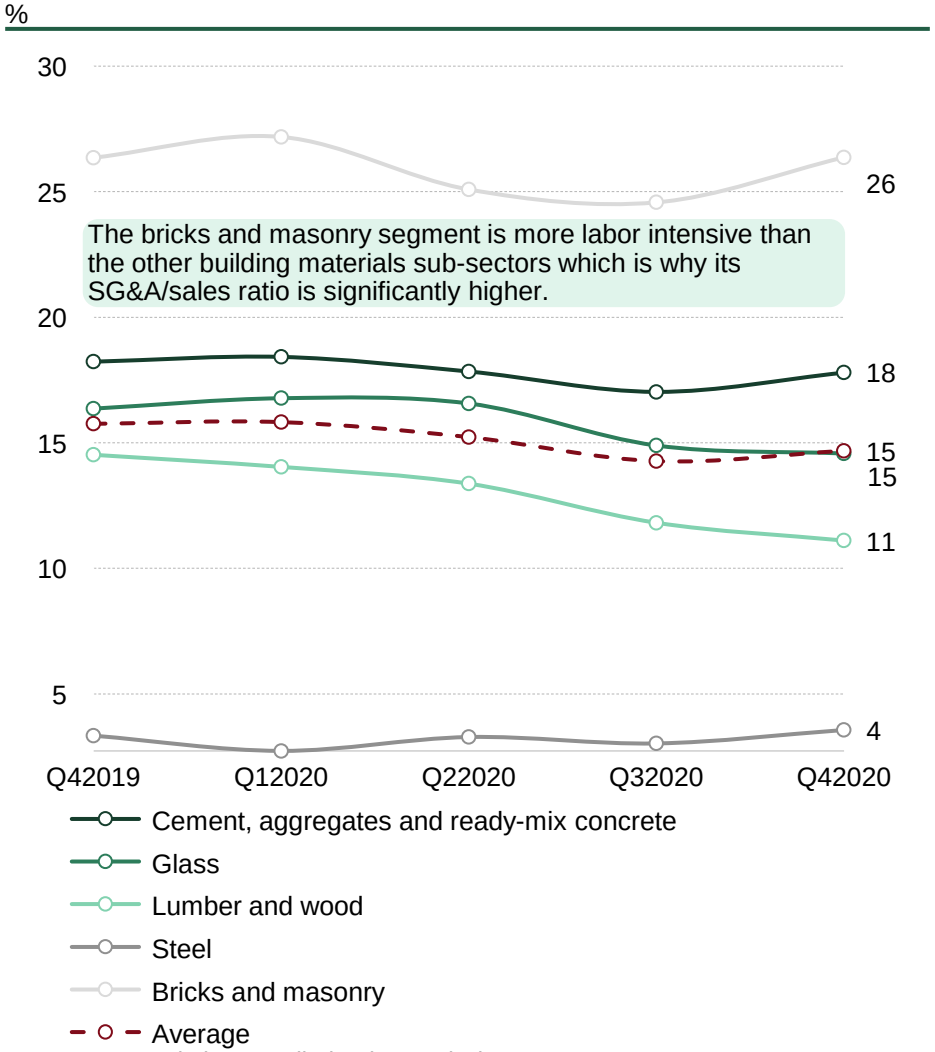




Source: Capital IQ, Applied Value analysis
Note: Arcelor Mittal and Heidelberg Cement data is based on half-year reporting

Lockdowns and furloughed employees during the pandemic have caused volatility in building products companies' SG&A over sales ratios

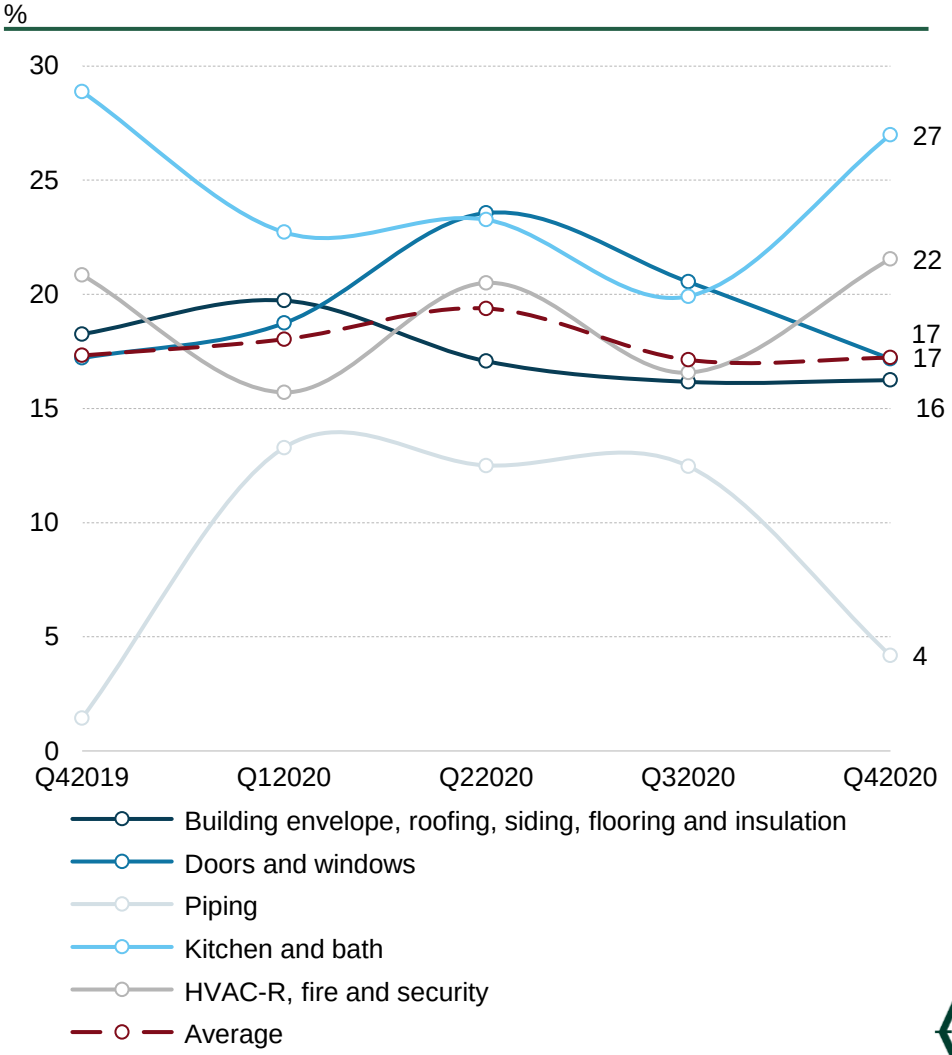
Building materials quarterly SG&A/sales development



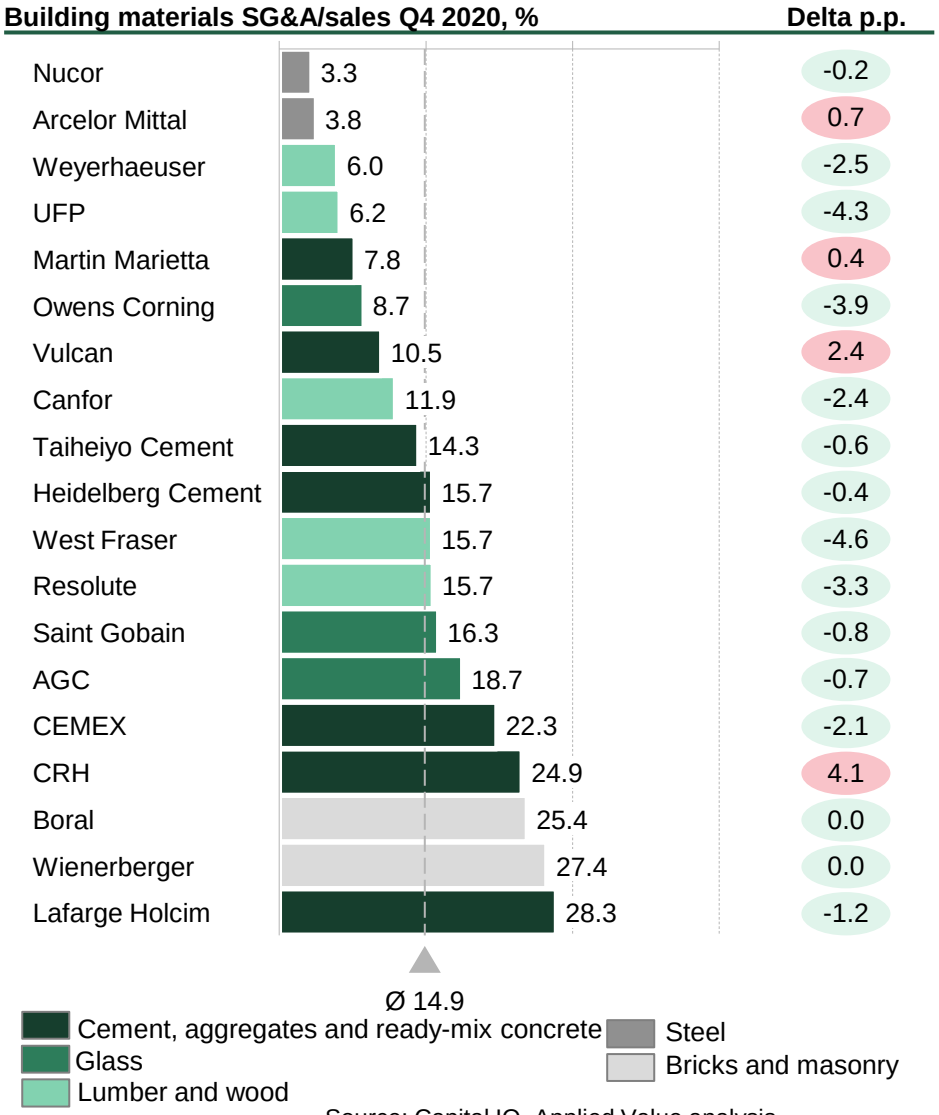
The bricks and masonry segment is more labor intensive than the other building materials sub-sectors which is why its SG&A/sales ratio is significantly higher.

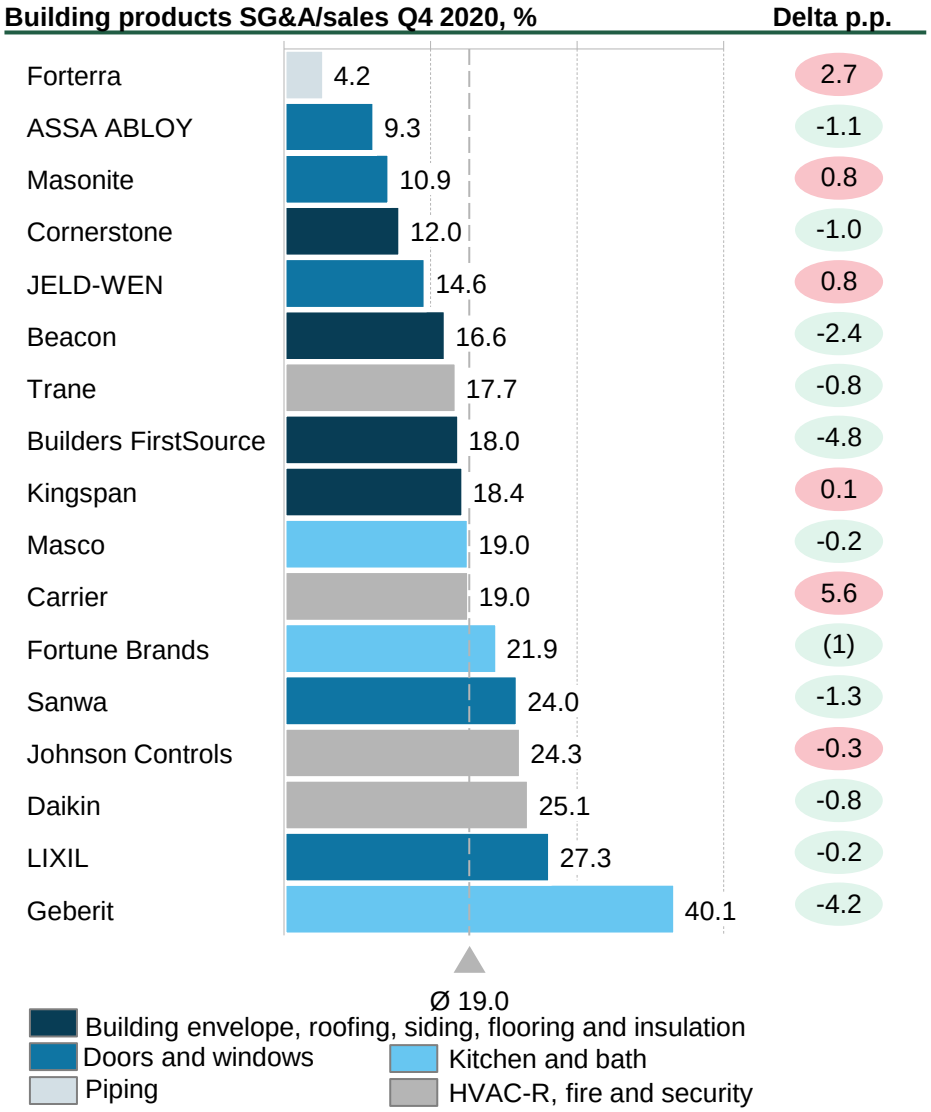
Source: Capital IQ, Applied Value analysis

Building products quarterly SG&A/sales development



A majority of the industry players managed to control SG&A over sales, Geberit stands out with a SG&A to sales ratio of 40%

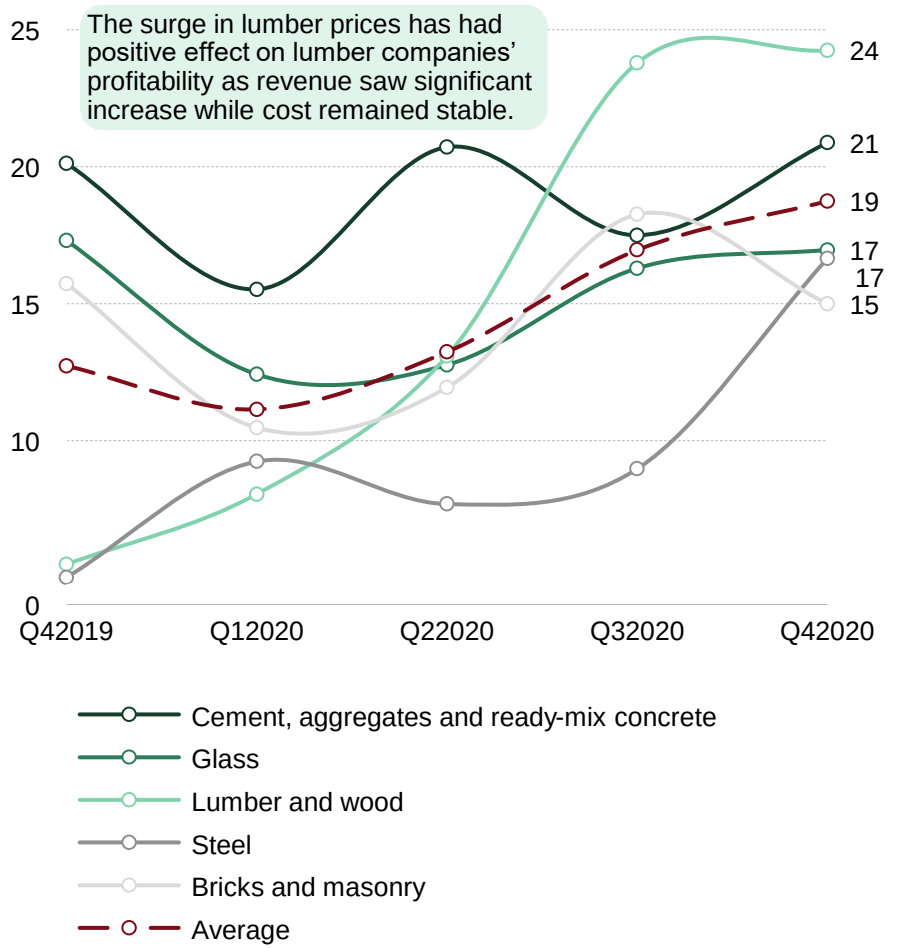




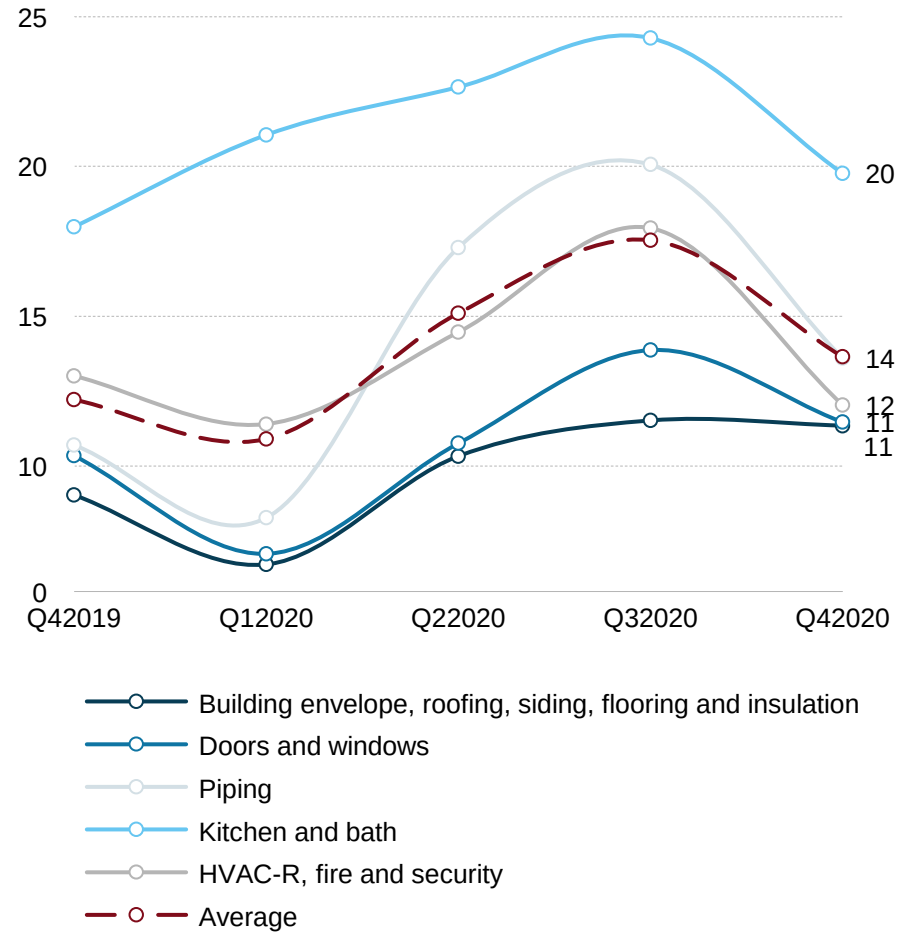
Source: Capital IQ, Applied Value analysis
Note: Arcelor Mittal and Heidelberg Cement data is based on data reported for H2

Lumber and steel companies have seen the biggest improvement in profitability YoY thanks to surging raw materials prices

Building materials EBITDA margin quarterly development
%

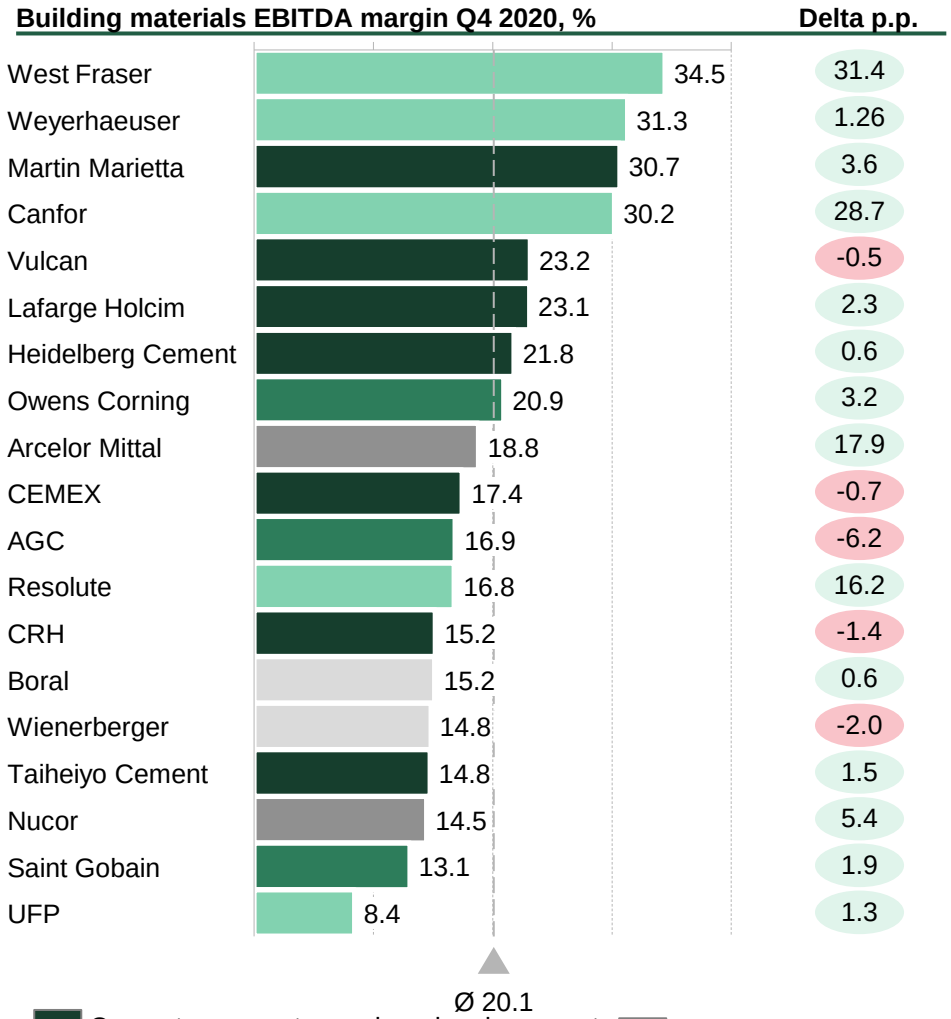


Building products EBITDA margin quarterly development
%

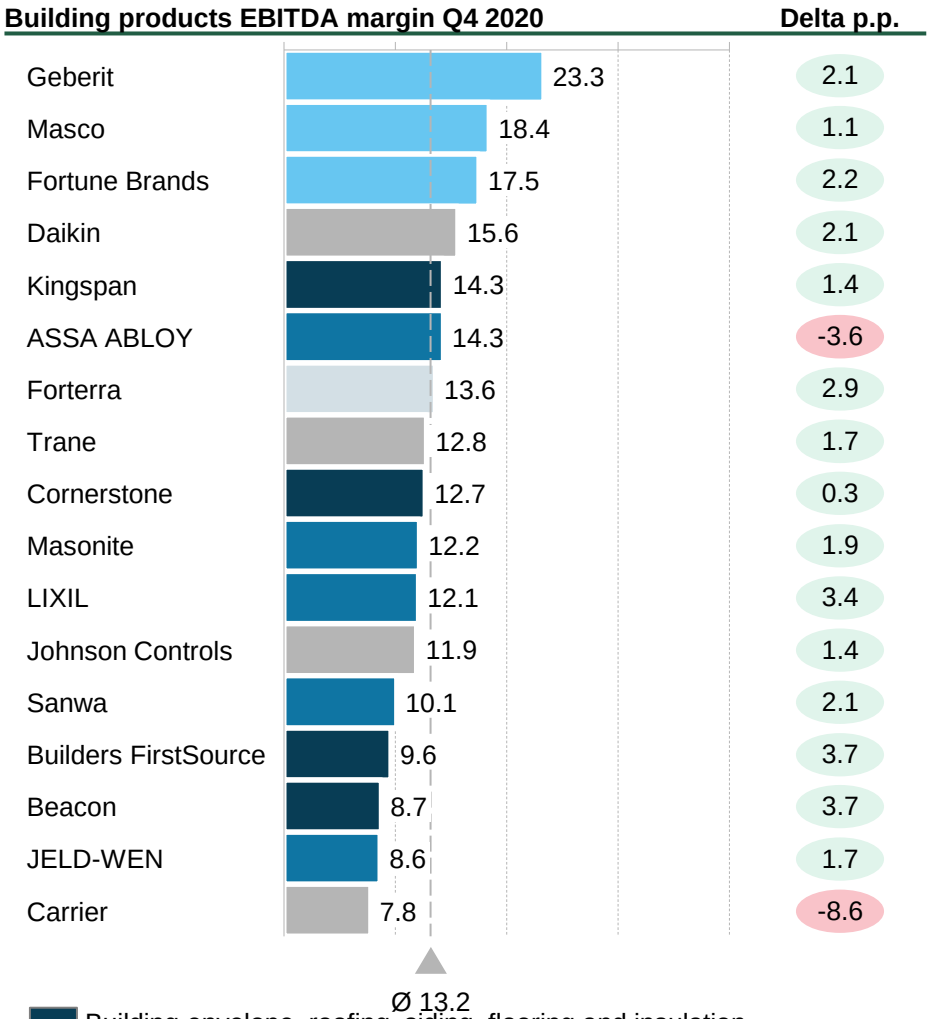


Source: Capital IQ, Applied Value analysis

Building materials have seen more volatility in profitability than building products due to large movements in the underlying raw materials markets

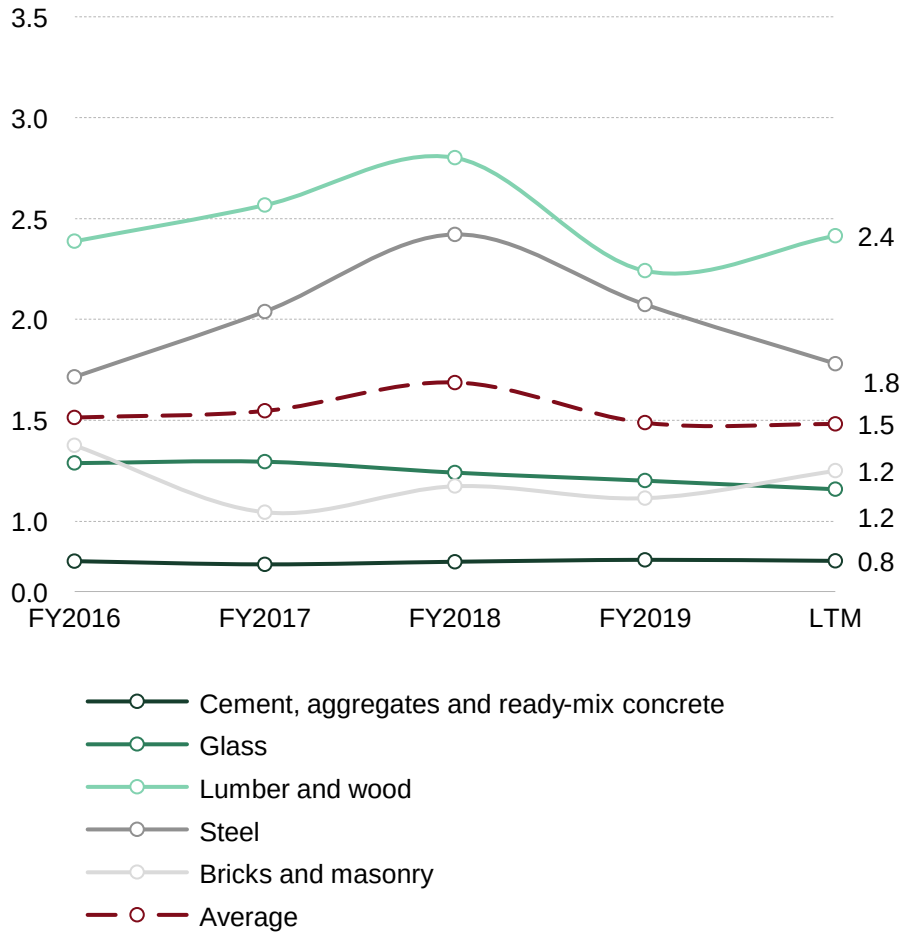


Source: Capital IQ, Applied Value analysis

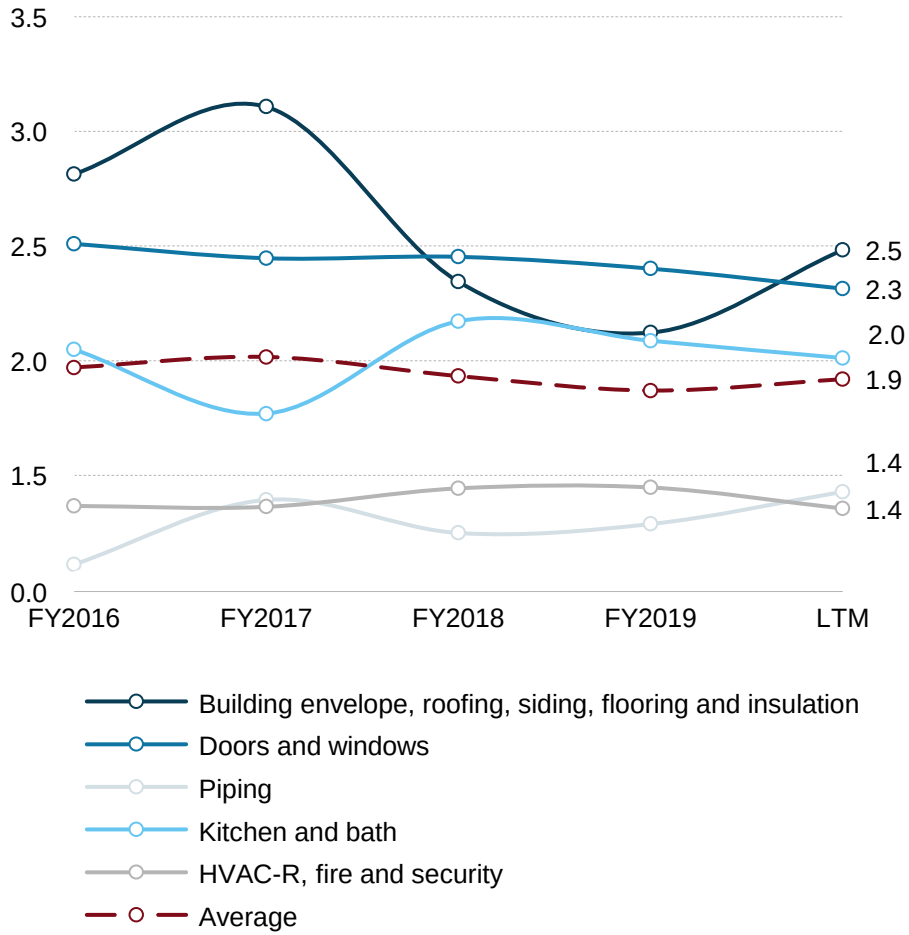


Lumber and building envelope companies saw an increase in FAT due to strong revenue development following the surge in lumber prices

Building materials fixed asset turnover yearly development

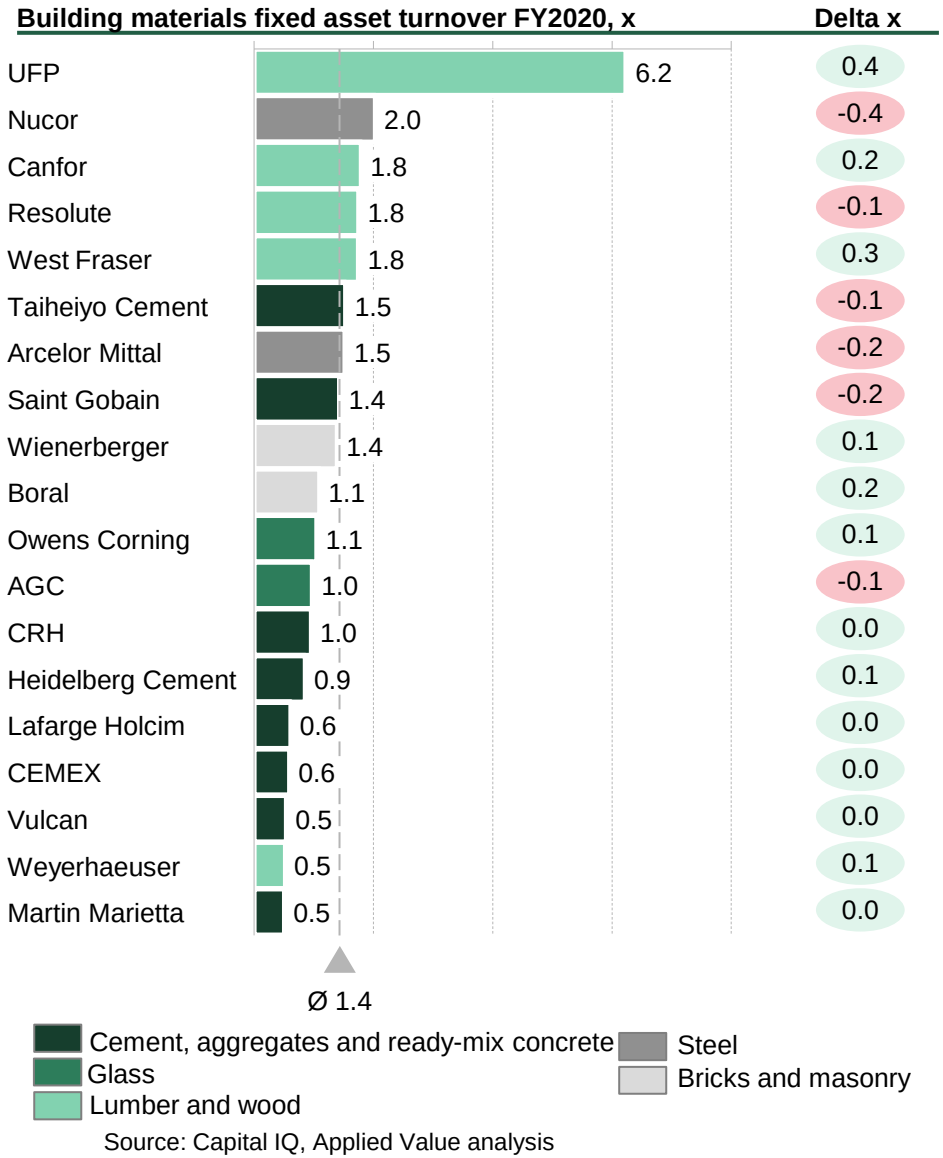


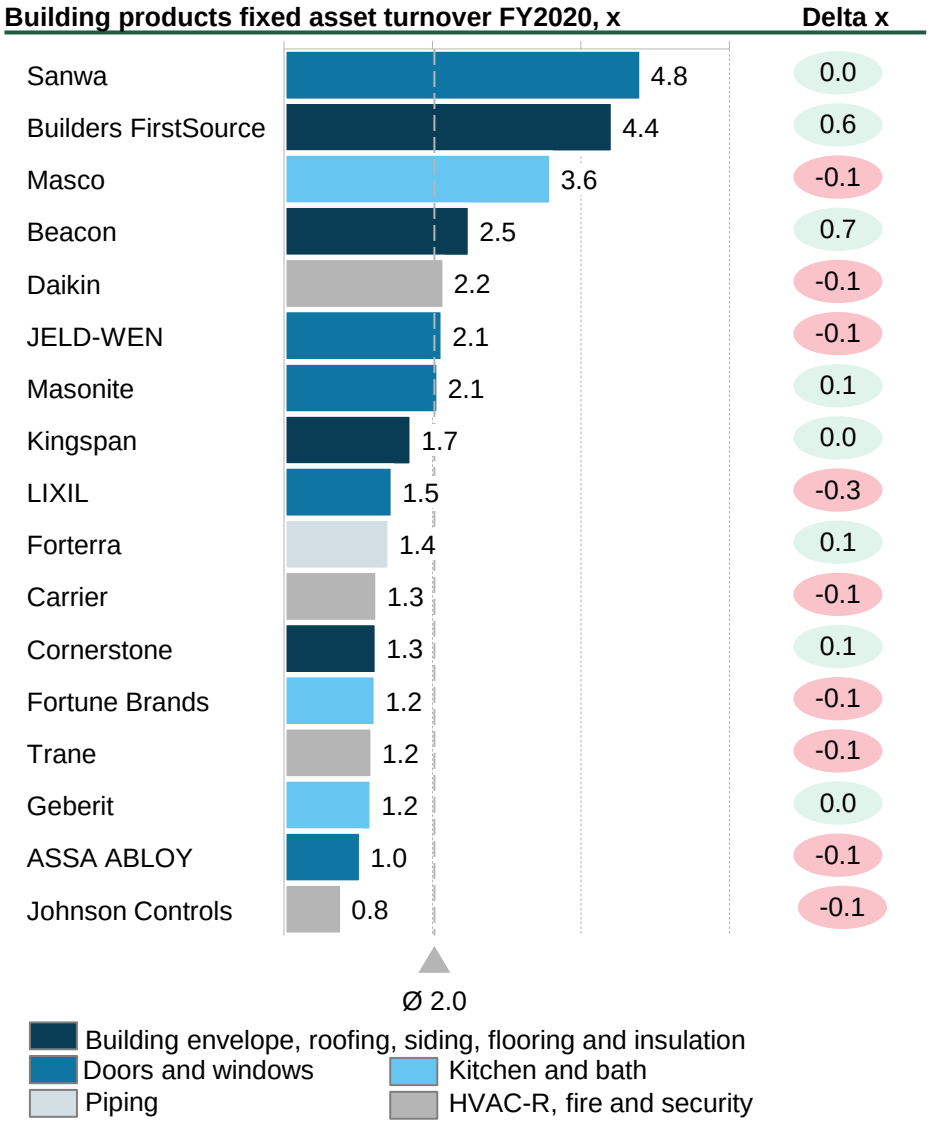
Building products fixed asset turnover yearly development



Source: Capital IQ, Applied Value analysis

Building products have on average higher FAT ratio than building materials companies since the sector is less capital intensive

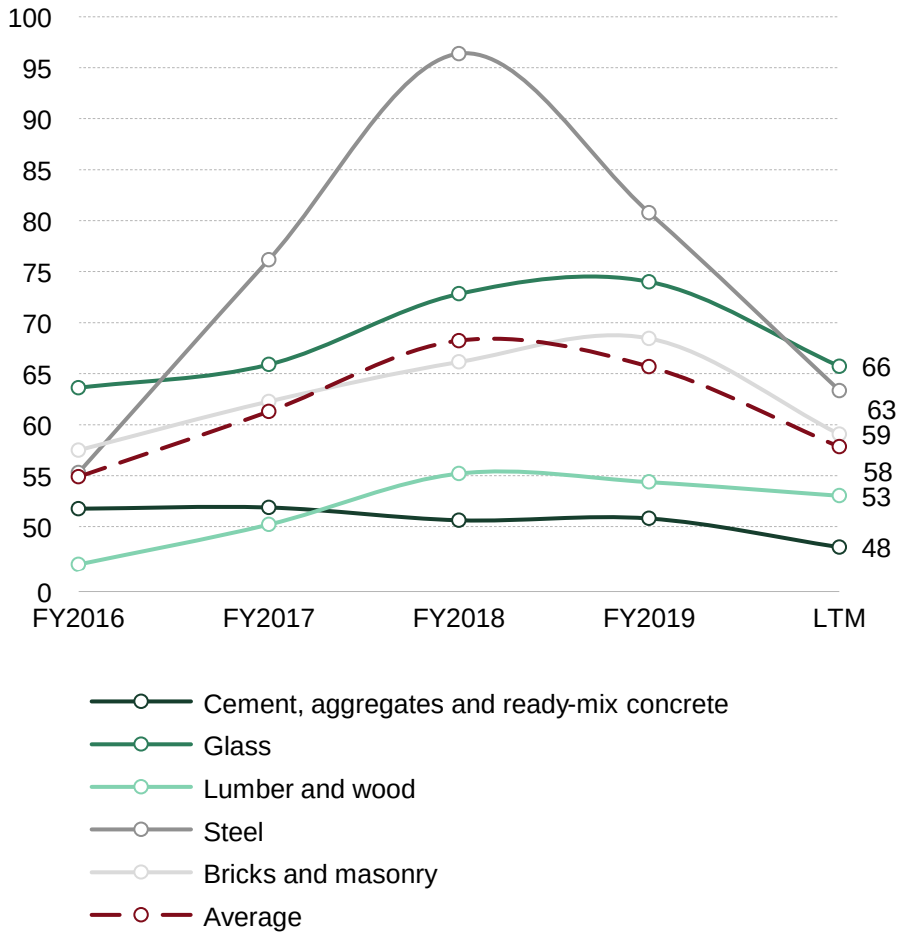




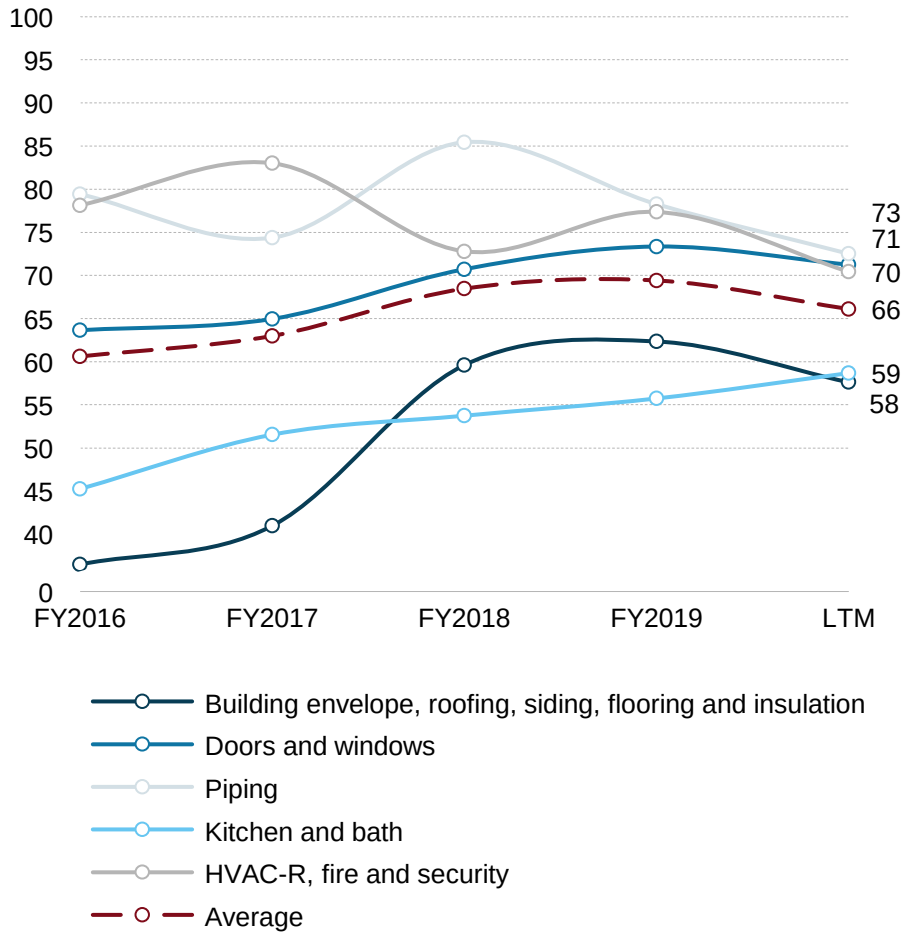
Source: Capital IQ, Applied Value analysis

The high demand for building materials and products LTM has led to improved working capital cycles for all analyzed sub-sectors

Building materials working capital cycle yearly development
Days

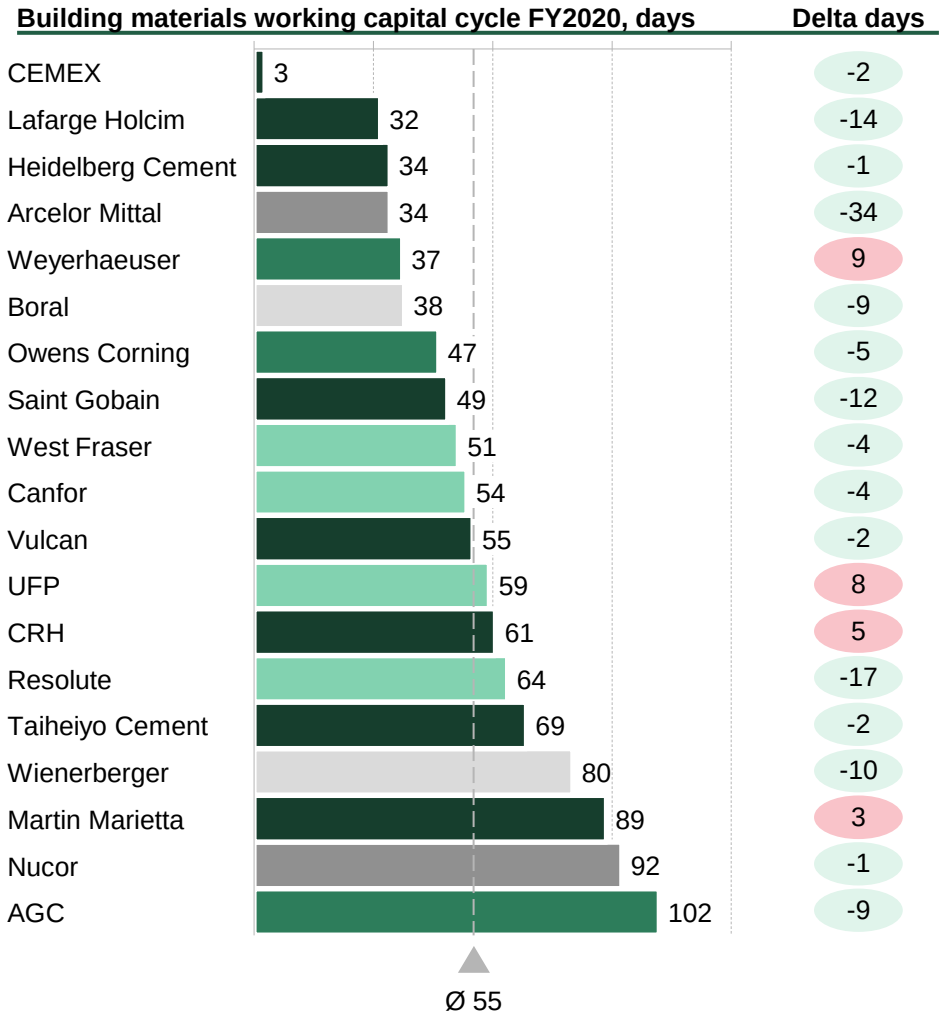


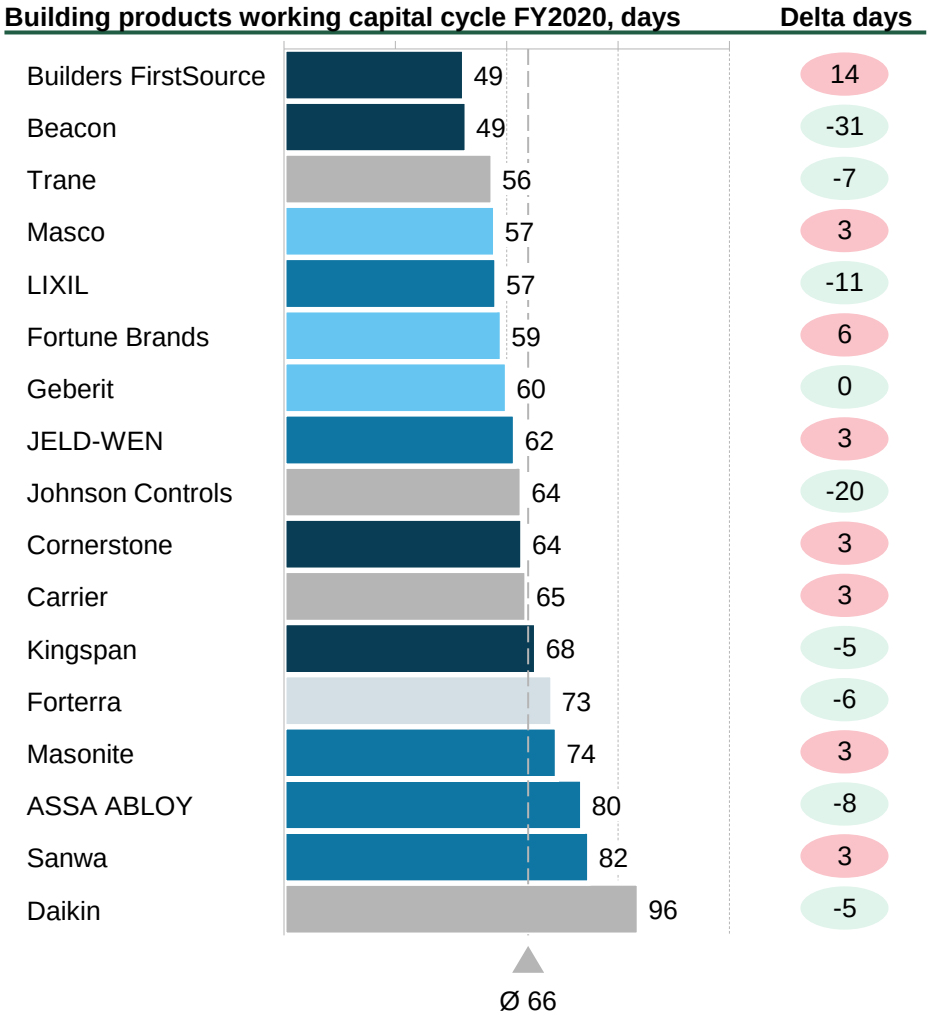
Building products working capital cycle yearly development
Days



Source: Capital IQ, Applied Value analysis

The benchmarked companies have seen significant improvement in working capital cycle, with Arcelor Mittal and Beacon seeing the largest effect

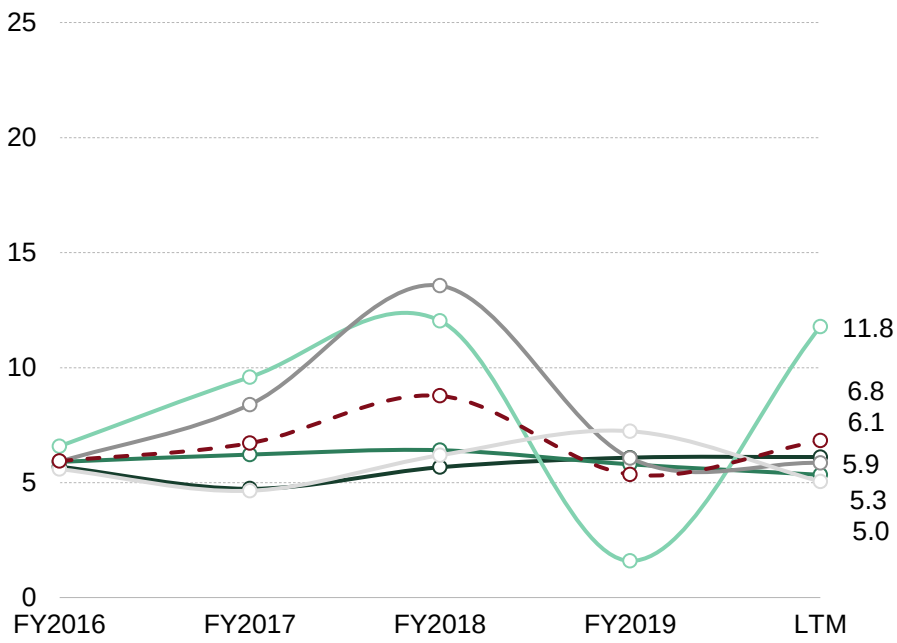




Source: Capital IQ, Applied Value analysis

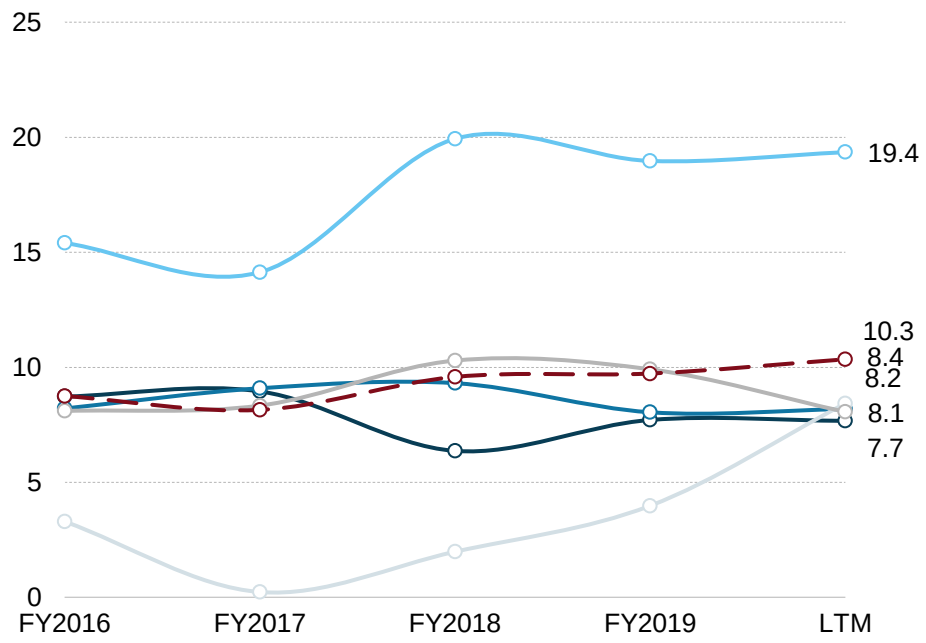
Kitchen and bath have outperformed other sub-sectors on ROIC L5Y, Lumber ROIC LTM has been driven by surging lumber prices during H2 2020

Building materials ROIC
%



- Cement, aggregates and ready-mix concrete
- Glass
- Lumber and wood
- Steel
- Bricks and masonry
- Average

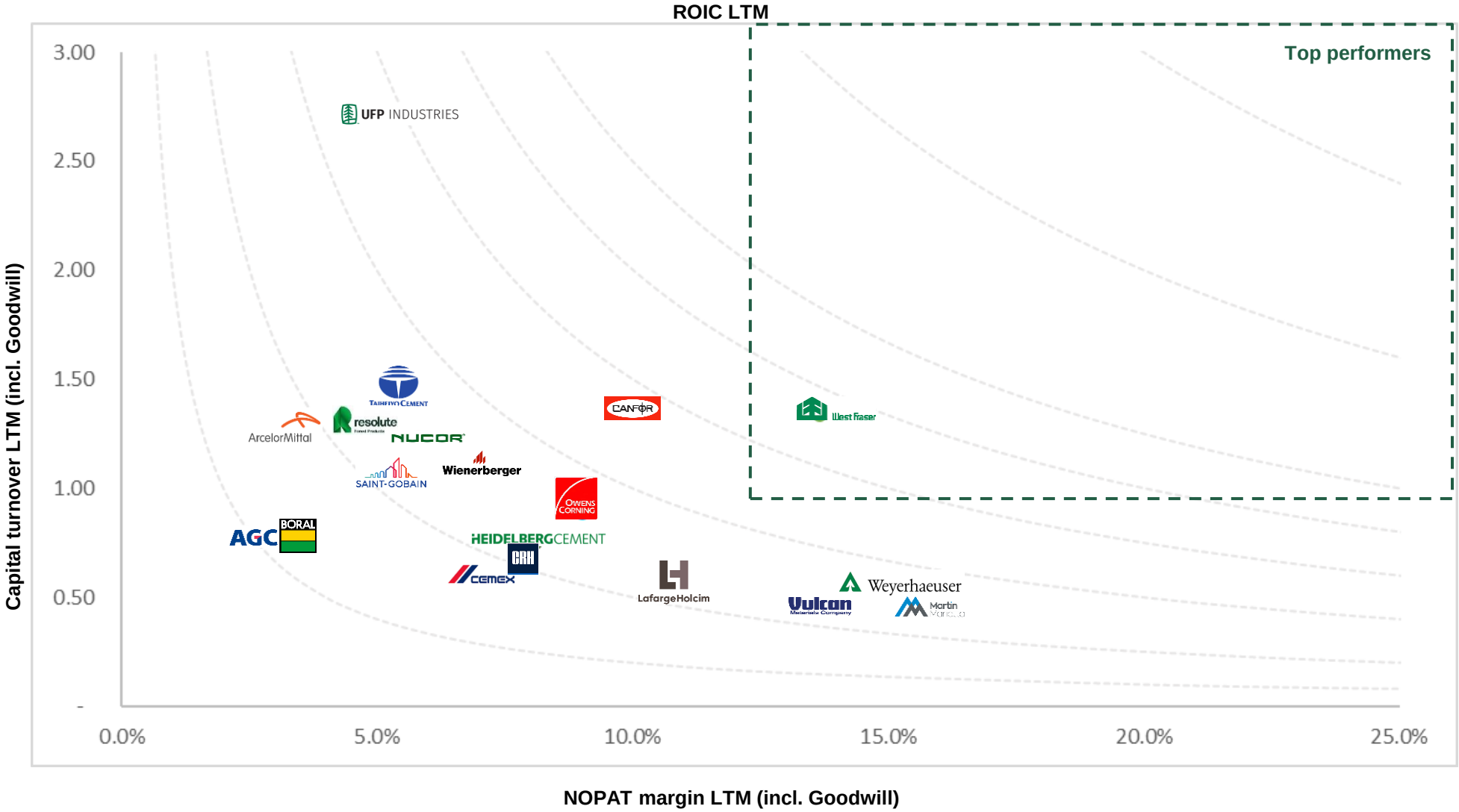
Building products ROIC
%



- Building envelope, roofing, siding, flooring and insulation
- Doors and windows
- Piping
- Kitchen and bath
- HVAC-R, fire and security
- Average

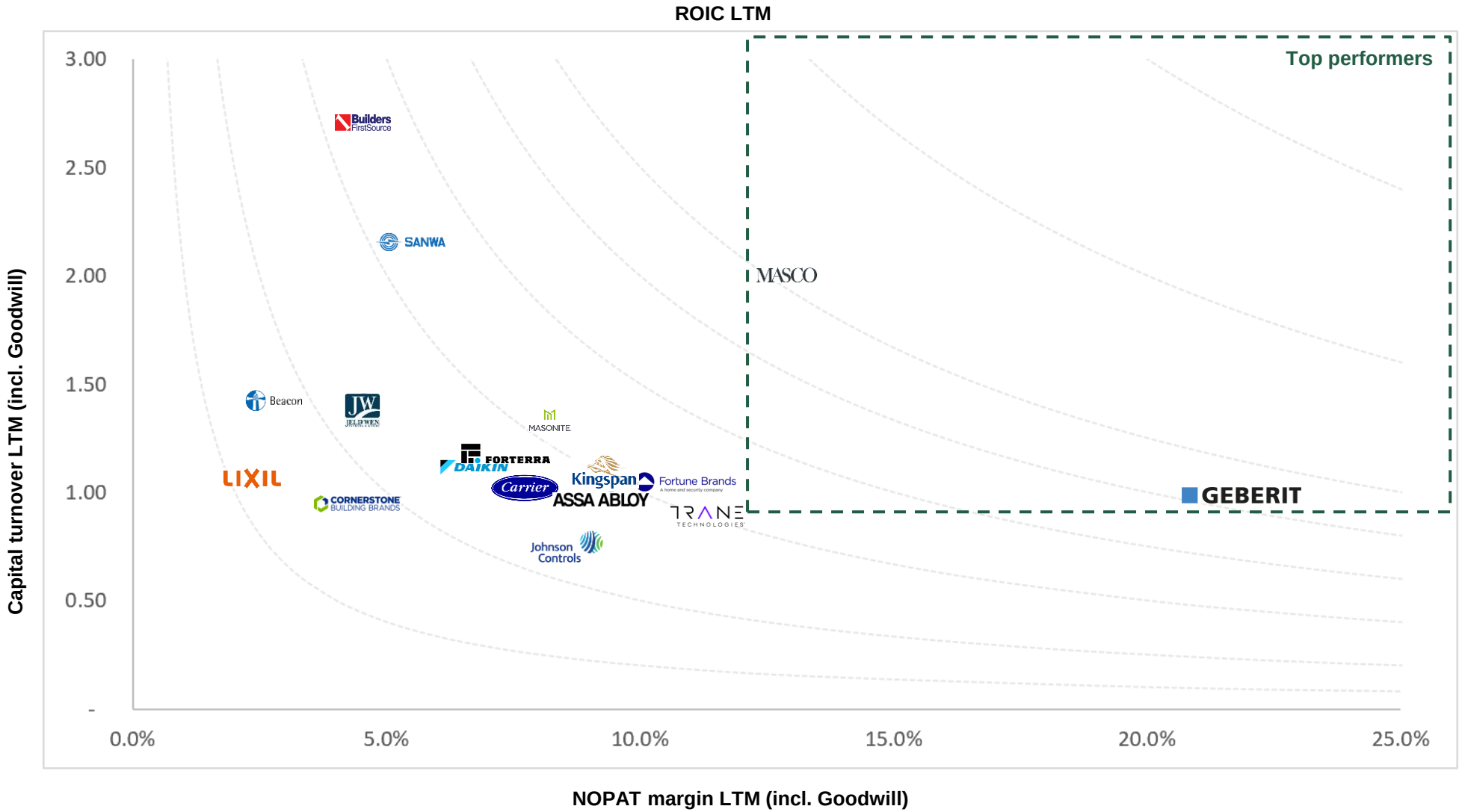
Source: Capital IQ, Applied Value analysis

Building materials ROIC performance varied between ~2% and ~18% LTM, with lumber company West Fraser as the top performer



Source: Capital IQ, Applied Value analysis

A majority of the benchmarked building products companies generated ~10% ROIC LTM, Masco and Geberit outperformed at 25.6% and 21.7%



Source: Capital IQ, Applied Value analysis

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Industry demand has primarily been driven by a pandemic induced boom in residential construction and renovation, and favorable interest rates

Tailwinds

Interest rate environment

- Mortgage rates have been falling during the last year and are at a record low

Housing market

- The outlook for the US housing market is strong, and is better than prior to the pandemic
- US residential real estate prices have been rising since June. The S&P CoreLogic Case-Shiller US National Home Price NSA Index was up 10.4% at year-end 2020

Renovation boom

- Home renovations and DIY home projects have been booming during the pandemic, leading to high demand for building products and materials

Raw materials price development

- Tight supply and strong demand for raw materials has led to significant price increase in the prices of lumber and steel, which has had positive effect on revenue development and profitability

Sources: S&P Dow Jones, National Association of Home Builders

Headwinds

Supply chain issues and delays

- The construction industry was hit hard at the beginning of the pandemic and the industry has been facing supply chain issues and materials shortages as a consequence
- A significant amount of construction projects have been delayed or stopped during Covid-19, which has negative effect on demand for building products and materials

Diminishing construction backlogs

- Construction backlogs have been decreasing throughout 2020, and were well below pre-pandemic levels at the end of Q4

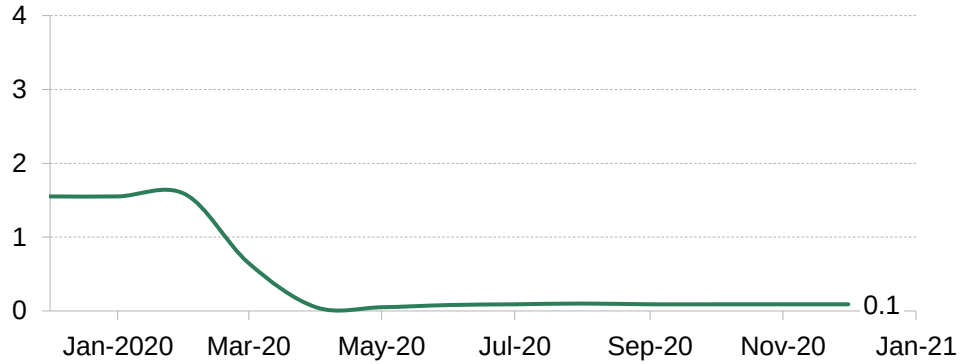
Sources: Associated Builders and Contractors, US Chamber of Commerce

The low interest rate environment is a key driver behind housing demand and home renovation projects

Interest Rates

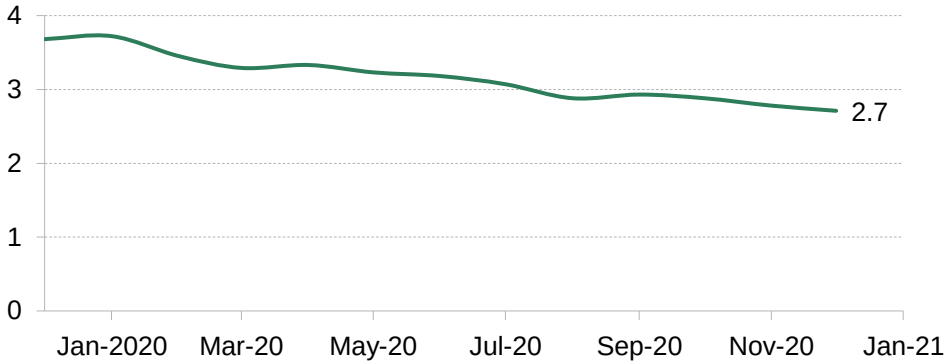
The FED rate has remained low throughout 2020 and US mortgage rates are at a record low, which has made owning a house more affordable

FED Rate
%



Source: FRED

30-Year Fixed Mortgage Rate
%

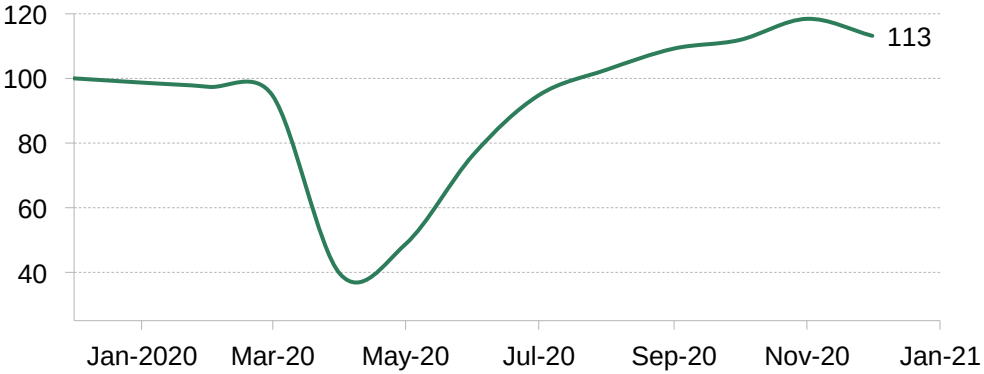


Source: Federal Home Loan Mortgage Corporation

Returning Confidence

The outlook for the housing market took a hit at the beginning of the pandemic. Housing market confidence has since seen strong recovery and the outlook for current and future home sales is now above pre-pandemic levels

Housing Market Index
Dec'19=100



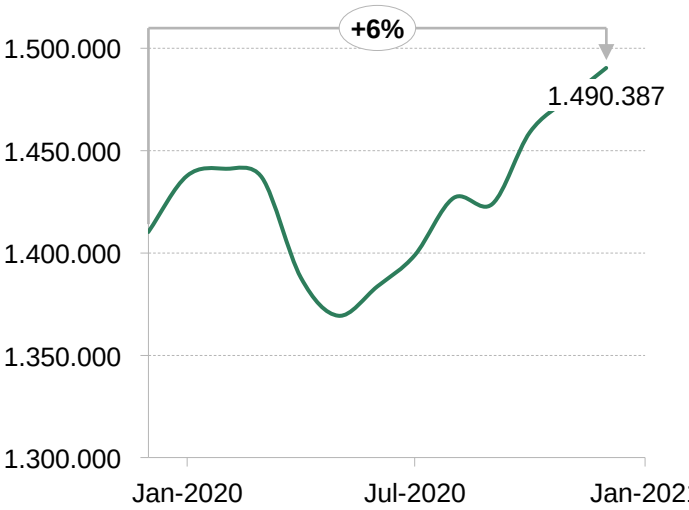
The National Association of Home Builders' Housing Market Index reflects builders' outlook for current home sales, home sales for the next six months and traffic of prospective buyers

Source: National Association of Home Builders

Confidence has returned to the market, construction industry spending reached its highest level of 2020 in Q4, and materials prices trended up

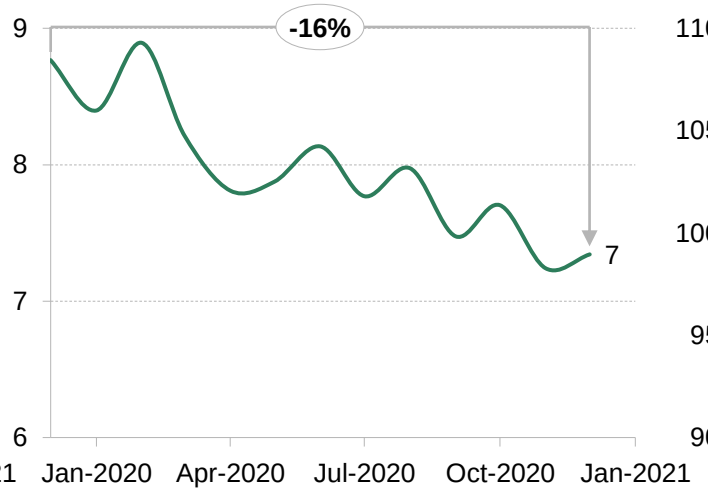
Construction spending	Diminishing backlog	Rising raw materials prices
The construction industry has seen increased spending during H2 2020, which has positive effect for building materials and products companies.	Backlogs have been decreasing since the start of the pandemic. Diminishing backlogs might have negative impact on future demand.	Prices have seen an upward spike during Q4. Sawmills and other production facilities shut down during lockdown, and supply has not been able to keep up with the rapidly rising demand.

Construction industry spending
$ MM



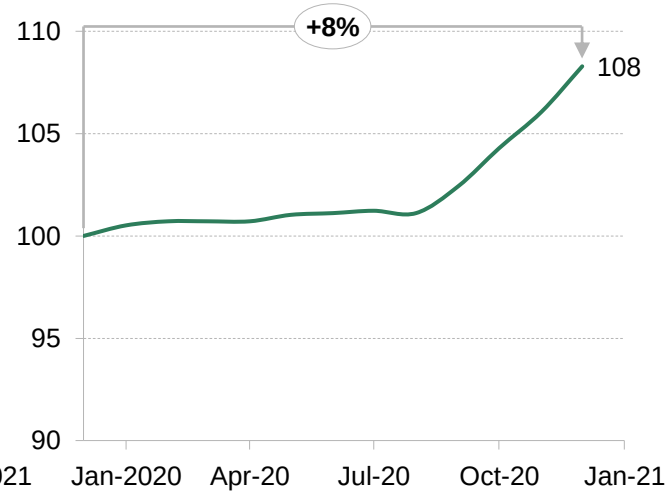
Source: Associated Builders and Contractors

Construction backlog indicator
Months



Source: Associated Builders and Contractors
Notes: Total months of forward-looking work under contract

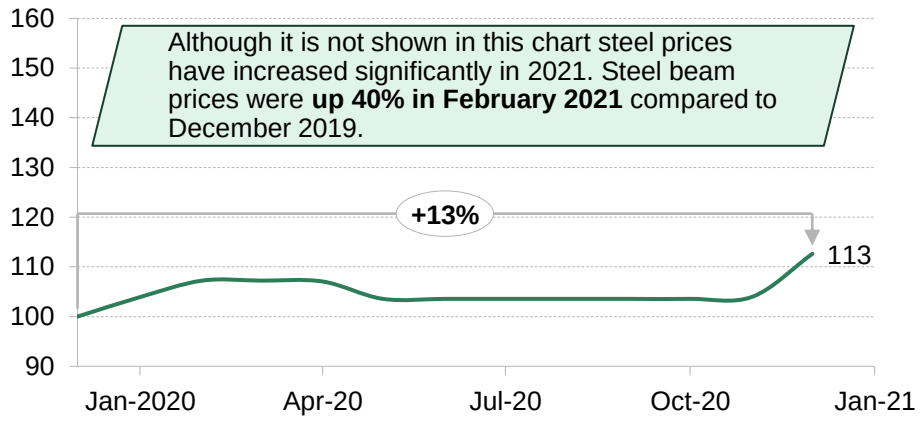
Materials price index
Dec'19=100



Source: ENR
Notes: Calculated as weighted price movement of structural steel, Portland cement, and 2x4 lumber.

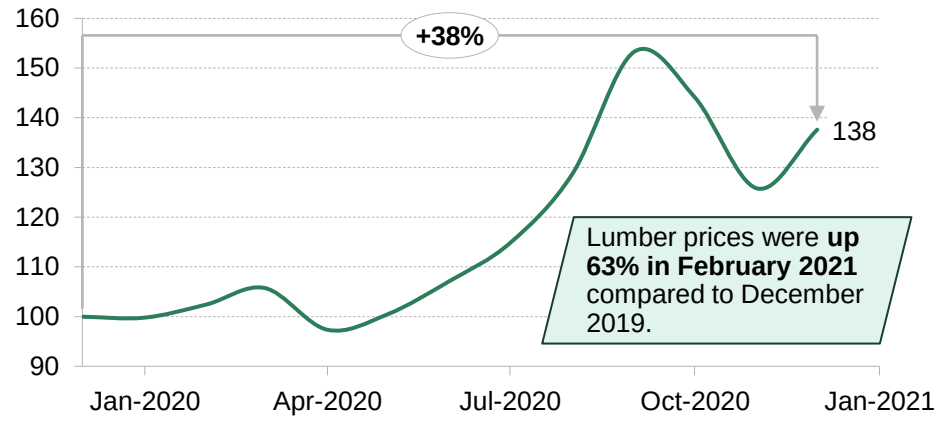
Lumber has seen a significant price increase during 2020 driven by tight supply and high demand, steel and glass saw an upward turn in Q4

Steel beam price (8x8 inch.)
Indexed price, Dec'19=100



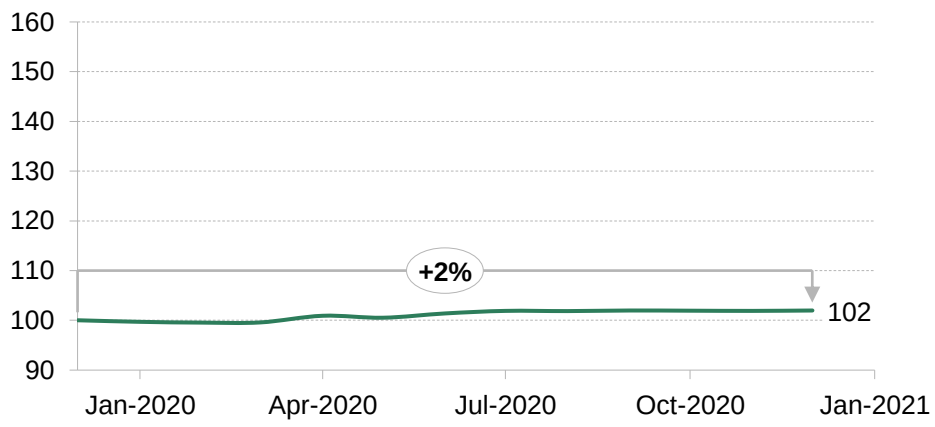
Source: AMM Fastmarket

Lumber price development
PPI, Dec'19=100



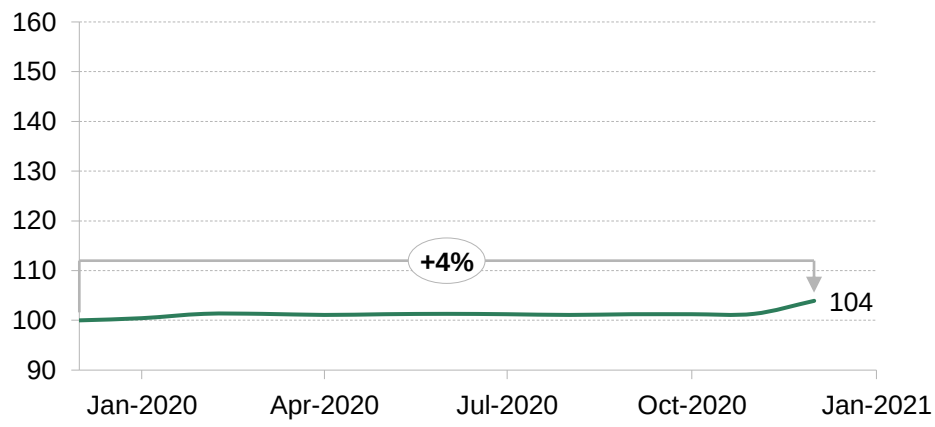
Source: FRED

Cement price development
PPI, Dec'19=100



Source: US Bureau of Labor Statistics

Flat glass price development
PPI, Dec'19=100



Source: FRED

M&A activity in the sector has been driven by horizontal integration and PE firms deploying dry powder

Recent M&A Transactions

Builders FirstSource → BMC	**Deal Value** 3.04 BUSD	The merger between Builders FirstSource and BMC will create a premium building products manufacturer and distributor with $11 BUSD in sales.
American Securities → FBM Foundation Building Materials	**Deal Value** 1.40 BUSD	Private Equity firm American Securities announced the acquisition of North America's largest specialty building products distributor Foundation Building Materials in late 2020.
Fortune Brands → Larson	**Deal Value** 0.74 BUSD	Fortune Brands expands its outdoor living segment with the acquisition of Larson and opens the door for Fortune Brands to new segments within the building products industry.

M&A Drivers

Industry characteristics	Consolidation	Private equity interest
The industry is fragmented, and growth is modest. This can be explained by the maturity of the industry as well as by commoditization of many building materials and products.	Horizontal integration has been a driver of industry transactions, as organic growth is hard to achieve in the fragmented industry. M&A has also been used to expand the acquirer's product portfolio.	PE firms' interest in the industry has seen an upswing recently and building products and materials have seen significant inflow of PE capital during H2 2020.

Sources: Capital IQ, MDM, Applied Value Analysis

Applied Value Building Products Report

Executive Summary & Introduction

Financial Benchmarking

Industry Trends

Applied Value Presentation

Applied Value is a management consulting & investment firm founded on the principles of Lean Growth and Entrepreneurship.

1997 Background as an internal consultancy and change agent within the Swedish ***Stenbeck Group…***

KORSNÄS · MTG · TELE2 · MILLICOM

2021 …today, a hands-on consultancy supporting clients **across industries**

Industries of Expertise

- **Sourcing & Procurement**
- **Supply Chain & Operations**
- **Commercial Excellence & Pricing**
- **Product & Innovation**
- **Strategy & Organization**
- **M&A**

Applied Value Group

Investments

Management Consulting

Social Responsibility

Selected portfolio companies

HUMAN CARE · COSMOSID · rivermeadow · Bake My Day · PNP · SÖDERFORS STEEL · Division 5 Inc STEEL FABRICATION

GM · PEPSICO · ERICSSON · MASONITE · ASSA ABLOY · ups · FRESENIUS MEDICAL CARE · SKANSKA

HAND IN HAND · Straight Ahead MINISTRIES · Exodus

ROI driven · *Fact-based* · *Practical over theoretical* · *Hands-On* · *Global perspectives*

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Applied Value challenges and supports repeat global clients across industries from six offices

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Our Lean Growth framework is based on Focus, Simplicity, Speed – guiding principles to raise client performance

Lean Growth

Focus

- ✓ Define strategic focus
- ✓ Excel in key capabilities
- ✓ Use partners for non-core activities

Simplicity

- ✓ Reduce complexity
- ✓ Simplify and standardize processes, organization and supply chain

Speed

- ✓ Set targets
- ✓ Measure and improve
- ✓ Reduce Market-Customer-Cash Time

Example Offerings

Strategy & Organization
Go-to-Market Strategy, G&A Productivity, Working Capital and CAPEX Efficiency, Digitalization & IT

Commercial Excellence & Pricing
True Profitability, Portfolio Optimization, Sales Efficiency, Pricing Model, Governance

Sourcing & Procurement
Sourcing Strategy, Direct & Indirect Materials and Services, Risk Management, SRM, Competence Development

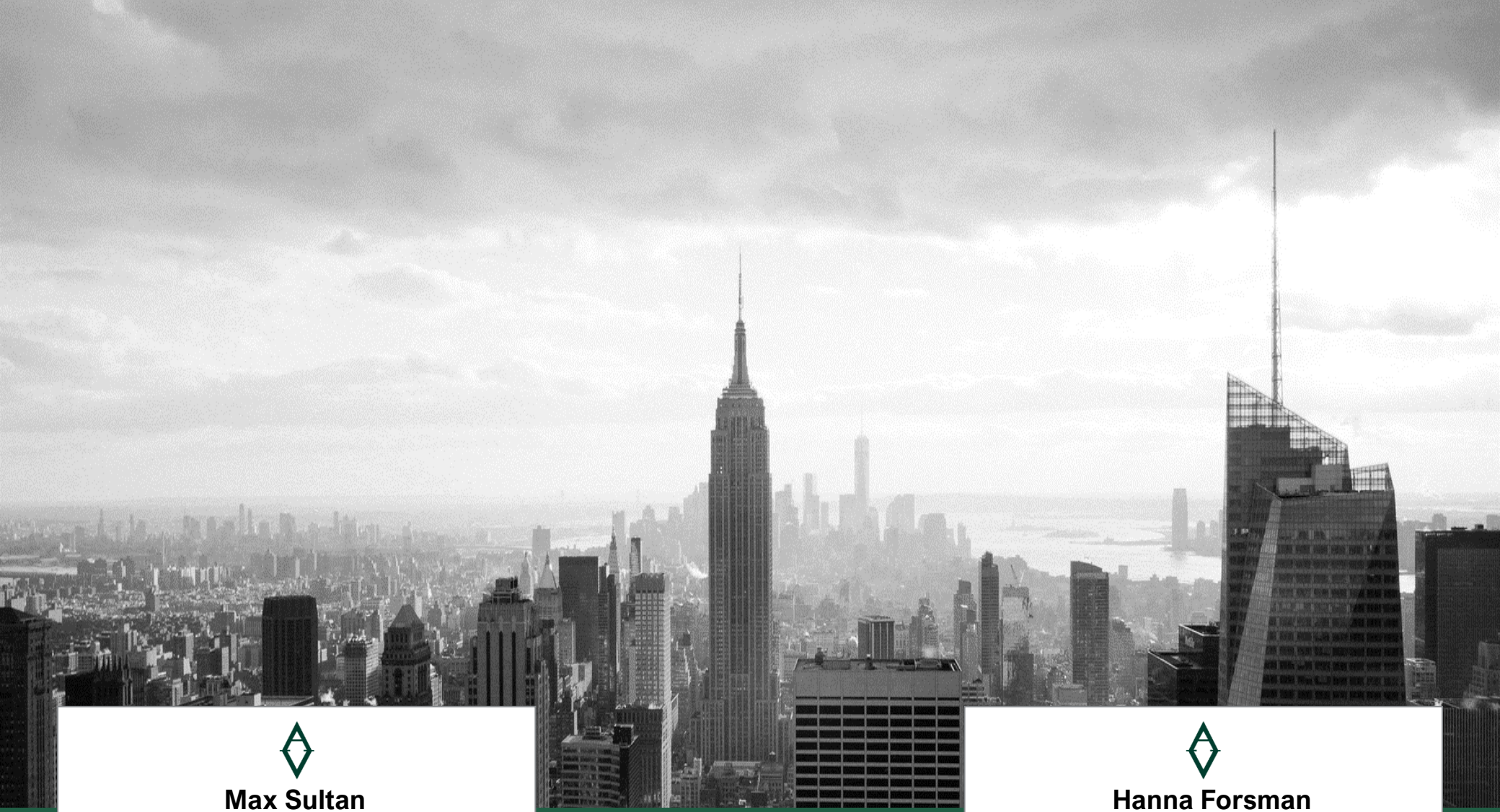
Supply Chain & Operations
Supply Chain Strategy, Inventory Management, Mfg. Footprint, S&OP, Logistics & Fulfillment, Order-to-Cash

Product & Innovation
Product Line Profitability, VA/VE, R&D Strategy & Efficiency, Product Lifecycle Management

Mergers & Acquisitions
Commercial Due Diligence, Operational Due Diligence, Post-Merger Integration

Applied Value has strong expertise in several areas relevant for driving improvements in the building materials and products industry

Applied Value Expertise	Improvement levers	Typical results
Commercial Excellence	› True Profitability & Product Portfolio Productivity › Pricing Strategy, Tools, and Governance › Salesforce Incentives and Training	› Portfolio rationalization by 20-60% › Price improvements of 5-10%
Sourcing & Supply Chain	› Direct & Indirect Commodity Strategy › Value Analysis / Value Engineering › Competence Development Programs	› 4-12% reduction of COGS in long-term Sourcing Programs › 5-25% reduction of spend in selected categories (mainly indirect material)
Manufacturing & Operations	› Manufacturing Footprint Strategy › Make v. Buy Decisions › Plant Overhead Reductions	› Reduction of 15-25% in mfg. network burden costs › Reductions of 20-30% in plant indirect
Order to Delivery	› Current State Process Mapping › Elimination of low-value add activities › Streamline Ways-of-Working	› “Order-to-Cash” lead time reductions of 15-45%
SG&A Efficiency	› Organizational Design › Headcount Drivers and Org. Dimensioning › Process Automation and Outsourcing	› 20-30% reduction of SG&A costs



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